

PRIME POLYNOMIALS IN LEGENDRE INTERVALS OVER $\mathbb{F}_{2^k}[t]$, II: EXACT VARIANCES VIA QUADRATIC TOWERS AND SUPERSINGULAR ARTIN–SCHREIER CURVES

RUQING CHEN

ABSTRACT. Let $q = 2^k$ and let $\mathcal{I}_f = \{f^2 + s : \deg s \leq d\}$ be the Legendre interval of a monic $f \in \mathbb{F}_q[t]$ of degree d . Chen's Open Problem 8.2 asks why the variance of the prime-power count $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ deviates by a factor as large as $q^2/3$ from the Keating–Rudnick prediction. We resolve the case $d = 4$ completely and unconditionally: using the quadratic tower $\mathbb{F}_q \subset \mathbb{F}_{q^2} \subset \mathbb{F}_{q^4} \subset \mathbb{F}_{q^8}$ we obtain a closed formula for Ψ on every class of the Hayes group $\mathcal{G}_3$, from which $\text{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$, $\text{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$, the full structure of the quadratic-twist correlations, and an elementary proof of Carlitz's cubic evaluation over $\mathbb{F}_{2^{8k}}$ follow. For $d = 6$ we prove that the Legendre profile takes exactly three values governed by $[c_2 = 0]$ and $\text{Tr}(c_4/c_2^2)$, reducing the variance to three exponential sums A_6, X, Y on $U = \{\alpha \in \mathbb{F}_{q^{12}} : \text{Tr } \alpha = \text{Tr } \alpha^3 = \text{Tr } \alpha^5 = 0\}$. We show that $A_6 - q^9$ and X are sums of Frobenius traces of supersingular Artin–Schreier curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ and reduce them to $\mathbb{F}_2$ -quadratic forms whose radicals are kernels of explicit linearised polynomials; the pure cubic and pure quintic parts are evaluated for all k , and for odd k the mixed part is evaluated completely: it is governed by the factorisation type of the Gold-type quintic $v^5 + v + c$ over $\mathbb{F}_q$, a $\mathbb{Z}/4$ -condition on the Frobenius, and the signs are fixed through the L -polynomials of the curves. This proves $A_6 = q^9 + (q-1)q^5$ for all odd k . The conjectured closed forms $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$, $X = (-1)^{k+1}q^6 + q^5$ in the remaining cases are verified exactly for $q \leq 128$ by an algorithm independent of the explicit formula. The third constant $Y = q^3 m_k$ is a twisted Frobenius trace on a fixed nine-dimensional complete intersection $X = V(e_1, e_3, e_5) \subset \mathbb{A}_{\mathbb{F}_2}^{12}$ (geometrically integral, normal, with explicit two-dimensional singular locus) and is not a character sum; we discuss its weight and rank. Consequently the Legendre variance at $d = 6$ obeys $\text{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$, not q^4 .

CONTENTS

1. Introduction	2
Main results	2
Methods	3
Reproducibility	3
2. Preliminaries	3
2.1. Hayes group, fibres, Legendre intervals	3
2.2. Ψ as a root count	3
2.3. Variances and quadratic twists	3
2.4. Quadratic characters are Artin–Schreier characters	4
3. The case $d = 4$: unconditional theorems	4
3.1. The quadratic tower	4
3.2. The closed formula	4
3.3. Cubic Artin–Schreier sums over $\mathbb{F}_{q^8}$	5
3.4. The full-group variance	6
4. The case $d = 6$: the three-value structure theorem	7
5. The quadratic tower $\mathbb{F}_{q^{12}} \supset \mathbb{F}_{q^6}$	8
6. Quadratic forms and supersingular Artin–Schreier curves	9
6.1. A_6 and X as sums of Frobenius traces	9
6.2. The pure cubic part	10

Date: October 8, 2026.

Preprint. Zenodo DOI: [10.5281/zenodo.23224897](https://doi.org/10.5281/zenodo.23224897). Code and data: <https://github.com/Ruqing1963/legendre-intervals-part2>.

6.3. The pure quintic part	10
6.4. Exact evaluation and the conjecture	11
6.5. The mixed part for odd k : a Gold-type quintic and a $\mathbb{Z}/4$ condition	12
6.6. The mixed part for odd k : the theorem	14
7. The scaling law at $d = 6$ and the nature of Y	16
7.1. Variance formula	16
7.2. The geometry behind Y : a fixed nine-dimensional variety and a twisted Frobenius	18
8. Summary of the numerical evidence	20
References	20

1. INTRODUCTION

Let $q = 2^k$, $d \geq 2$, $n = 2d$ and $\ell = d - 1$. For a monic $f \in \mathbb{F}_q[t]$ of degree d the *Legendre interval* is

$$\mathcal{I}_f = \{f^2 + s : s \in \mathbb{F}_q[t], \deg s \leq d\}, \quad |\mathcal{I}_f| = q^{d+1},$$

and $\Psi(f) = \sum_{P \in \mathcal{I}_f} \Lambda(P)$ counts prime powers in it with the von Mangoldt weight. In odd characteristic the sets $\mathcal{I}_f$ are ordinary short intervals and the variance of Ψ is governed by the Keating–Rudnick theorem [12], which predicts $\text{Var}(\Psi) \sim (d-2)q^{d+1}$ as $q \rightarrow \infty$. In characteristic 2 the squaring map on the Hayes group is a homomorphism, the Legendre intervals are the fibres over the subgroup of squares, and Chen [4] observed (Table 4 of *op. cit.*) that the variance over Legendre intervals exceeds the full-group variance by factors $4/3$, $16/3$, $64/3$ for $q = 2, 4, 8$ at $d = 4$ and asked for an explanation (Open Problem 8.2).

This paper is the second part of our study. Part I established the dictionary between Legendre intervals and quadratic characters of the Hayes group and verified, by exact computation for $q \leq 8$, the identities $\text{Var}_{\mathcal{G}^2}(\Psi)/q^5 = q(q-1)$ and $\text{Var}_{\mathcal{G}}(\Psi)/q^5 = 3 - 3/q$. Here we prove them for all $q = 2^k$, and we carry the analysis to $d = 6$, where a genuinely new phenomenon appears: two of the three constants that control the Legendre variance are elementary while the third is a Frobenius trace that is not a character sum.

Main results. All variances below are second moments about the global mean q^{d+1} (this is the quantity tabulated in [4]); see Section 2.

- $d = 4$ (**Section 3**). Theorem 3.2 gives Ψ on every class of the Hayes group $\mathcal{G}_3$ in closed form. Consequently (Theorems 3.4, 3.11, 3.12) for every $q = 2^k$

$$\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1), \quad \frac{\text{Var}_{\mathcal{G}}(\Psi)}{q^5} = 3 - \frac{3}{q}, \quad \frac{\text{Var}_{\mathcal{G}^2}}{\text{Var}_{\mathcal{G}}} = \frac{q^2}{3},$$

the deviation $\Psi - q^5$ is supported on the subgroup $\{c_1 = 0\}$, the identity class carries the deficit $-(q^4 - q^3)$, and the quadratic-twist correlations $\rho(\varepsilon)$ equal 1 on the $q-1$ characters of conductor u^2 and $(q-3)/(3(q-1))$ on all others when k is odd.

- $d = 6$, **structure (Section 4)**. Theorem 4.2: on the q^2 Legendre classes $(0, c_2, 0, c_4, 0)$ the function Ψ takes exactly three values,

$$\Psi(0, c_2, 0, c_4, 0) = q^7 + \frac{A_6 - q^9 - X}{q^2} + \frac{X}{q}[c_2 = 0] + \frac{Y}{q}[c_2 \neq 0] \chi\left(\frac{c_4}{c_2^2}\right),$$

where $A_6 = |U|$, $X = \sum_U \chi(e_2)$, $Y = \sum_U \chi(e_4)$ and $U = \{\alpha \in \mathbb{F}_{q^{12}} : \text{Tr } \alpha = \text{Tr } \alpha^3 = \text{Tr } \alpha^5 = 0\}$. The proof uses only the $\mathbb{F}_q$ -translation and scaling symmetries of U .

- $d = 6$, **the constants (Sections 5–6)**. Theorem 5.3 reduces A_6, X, Y to explicit finite sums over $\mathbb{F}_{q^6}$. Theorem 6.1 expresses $A_6 - q^9$ and X through the Frobenius traces of the supersingular curves $y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$, equivalently through $\mathbb{F}_2$ -quadratic forms on $\mathbb{F}_{q^{12}}$ with explicit radicals. Lemmas 6.2 and 6.4 evaluate the pure cubic and pure quintic parts for all k . For odd k , Theorem 6.12 evaluates the mixed part: $W(\lambda_3, \lambda_5) = 2q^7$ or 0 according as the Frobenius permutes the roots of the Gold-type quintic $v^5 + v + c$, $c = (\lambda_5^3/\lambda_3^5)^{1/2}$, as a 4-cycle or not; with the exact count of 4-cycles (Proposition 6.8) this

gives $A_6 = q^9 + (q-1)q^5$ for all odd k (Corollary 6.13). Conjecture 6.5, verified exactly for $q \leq 128$ (Table 1), states $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$ and $X = (-1)^{k+1}q^6 + q^5$ in general.

- **Scaling law and Y (Section 7).** Theorem 7.1 gives $\text{Var}_{\mathcal{G}^2}(\Psi)/q^7$ at $d = 6$ in closed form in terms of (q, k, m_k) , $m_k = Y/q^3$; it implies $\text{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ (Table 3) and $\lim \text{Var}_{\mathcal{G}^2}/q^9 = 1$ provided $m_k = o(q^{5/2})$. We identify Y as the trace of a twisted Frobenius σF^k on the cohomology of a fixed nine-dimensional complete intersection $X \subset \mathbb{A}_{\mathbb{F}_2}^{12}$ with an Artin–Schreier sheaf (Propositions 7.4–7.5, Theorem 7.7) and show that the data $m_k = 13, -1, -35, 47, 733$ ($k \leq 5$) exclude a pure weight-3 object of rank ≤ 5 .

Methods. Two ideas carry the paper. The first is the *quadratic tower*: for $n = 2d$ the extension $\mathbb{F}_{q^n}/\mathbb{F}_{q^{n/2}}$ is quadratic, so every $\alpha \in \mathbb{F}_{q^n}$ is a root of $t^2 + at + n$ with $(a, n) \in \mathbb{F}_{q^{n/2}}^2$, and the elementary symmetric functions of the n conjugates of α become explicit polynomials in traces of (a, n) . Counting pairs (a, n) with the irreducibility condition $\text{Tr}_{q^{n/2}/2}(n/a^2) = 1$ turns the prime-counting problem into linear algebra over $\mathbb{F}_q$ and $\mathbb{F}_2$. The second is the use of the *symmetries of $U_d = \{p_j = 0, j \text{ odd} \leq \ell\}$* : U_d is stable under $\alpha \mapsto \alpha + c$ ($c \in \mathbb{F}_q$) and $\alpha \mapsto s\alpha$ ($s \in \mathbb{F}_q^\times$), and the elementary symmetric functions transform by binomial coefficients mod 2; this alone forces the shape of the Legendre profile.

Reproducibility. All statements labelled “verified” were checked with exact integer arithmetic; the scripts `legendre_var_check.py`, `legendre_d4_formula.py`, `legendre_q16_hayes.py`, `legendre_d6_tower.py`, `legendre_d6_qf.py`, `legendre_d6_qf_mixed.py`, `quintic_check.py` and `kasami_sign.py` accompany the paper (figures: `make_figures.py`, `generate_figures.py`). The Hayes-group FFT of [4] was used as an independent check for $q \leq 32$ at $d = 6$ and $q \leq 128$ at $d = 4$.

2. PRELIMINARIES

2.1. Hayes group, fibres, Legendre intervals. Let $\mathcal{G} = \mathcal{G}_\ell = \{1 + c_1u + \cdots + c_\ell u^\ell\} \subset (\mathbb{F}_q[u]/u^{\ell+1})^\times$, $|\mathcal{G}| = q^\ell$. A monic $F = t^n + a_{n-1}t^{n-1} + \cdots + a_0$ maps to $F^*(u) = u^n F(1/u) \bmod u^{\ell+1}$, i.e. to its top ℓ coefficients $(c_1, \dots, c_\ell) = (a_{n-1}, \dots, a_{n-\ell})$; the map is multiplicative and each fibre has $q^{n-\ell} = q^{d+1}$ elements.

Lemma 2.1. *The Legendre interval $\mathcal{I}_f$ is the fibre over the class of f^2 , and the classes of squares form the subgroup $\mathcal{G}^2 = \{g^2\}$, of index q^m with $m = \#\{j \leq \ell : j \text{ odd}\} = \lfloor d/2 \rfloor$. Hence $\Psi(f) = \Psi(g)$ depends only on $g = [f^2] \in \mathcal{G}^2$.*

Proof. $f^2 + s$ has vanishing coefficients in all odd degrees $> d$, and $(1 + \sum c_j u^j)^2 = 1 + \sum c_j^2 u^{2j}$. The fibre of $[f^2]$ is $\{f^2 + s : \deg s \leq d\}$ since the top $\ell = d-1$ coefficients of $f^2 + s$ are those of f^2 . $\square$

2.2. Ψ as a root count. For $\alpha \in \mathbb{F}_{q^n}$ write $\text{charpoly}(\alpha) = \prod_{i < n} (t - \alpha^{q^i})$. If $F = P^j$ with $\deg P = n/j$, exactly $n/j = \Lambda(F)$ elements α have $\text{charpoly}(\alpha) = F$. Hence

$$(1) \quad \Psi(g) = \sum_{F \in \text{fibre}(g)} \Lambda(F) = \#\{\alpha \in \mathbb{F}_{q^n} : (e_1, \dots, e_\ell)(\alpha) = g\},$$

where $e_j(\alpha)$ denotes the j -th elementary symmetric function of the n conjugates of α . In particular $\sum_g \Psi(g) = q^n$ and the mean over $\mathcal{G}$ is q^{d+1} .

2.3. Variances and quadratic twists. Put $\varphi = \Psi - q^{d+1}$ and

$$\text{Var}_{\mathcal{G}}(\Psi) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \varphi(g)^2, \quad \text{Var}_{\mathcal{G}^2}(\Psi) = \frac{1}{|\mathcal{G}^2|} \sum_{g \in \mathcal{G}^2} \varphi(g)^2.$$

For a character χ of $\mathcal{G}$ let $\psi_n(\chi) = \sum_{\deg F=n} \Lambda(F) \chi(F)$. Then $\varphi(g) = |\mathcal{G}|^{-1} \sum_{\chi \neq 1} \psi_n(\chi) \bar{\chi}(g)$, and for ε a character trivial on $\mathcal{G}^2$,

$$(2) \quad \rho(\varepsilon) := \frac{\sum_{\chi} \psi_n(\chi) \overline{\psi_n(\chi \varepsilon)}}{\sum_{\chi \neq 1} |\psi_n(\chi)|^2} = \frac{\sum_g \varphi(g)^2 \varepsilon(g)}{\sum_g \varphi(g)^2}, \quad \frac{\text{Var}_{\mathcal{G}^2}}{\text{Var}_{\mathcal{G}}} = 1 + \sum_{\varepsilon \neq 1, \varepsilon|_{\mathcal{G}^2}=1} \rho(\varepsilon).$$

For $\chi \neq 1$ the L -function $L(u, \chi) = \sum_F \chi(F) u^{\deg F}$ is a polynomial of degree $\leq \ell - 1 = d - 2$ with inverse roots of absolute value $\sqrt{q}$, and $\psi_n(\chi) = -\sum_j \gamma_j^n$. Thus $\text{Var}_{\mathcal{G}}(\Psi)/q^{d+1} = |\mathcal{G}|^{-1} \sum_{\chi \neq 1} |\text{Tr } \Theta_\chi^n|^2$ with $\Theta_\chi \in U(d-2)$ the normalised Frobenius; the Keating–Rudnick prediction is the Haar average $d-2$.

2.4. Quadratic characters are Artin–Schreier characters.

Proposition 2.2. *Let $p_j(g)$ be the j -th power sum of the “roots” of $g \in \mathcal{G}_\ell$, defined by Newton’s identities (in characteristic 2: $p_j = \sum_{i < j} c_i p_{j-i} + [j \text{ odd}] c_j$). Then $p_j(gh) = p_j(g) + p_j(h)$, $p_{2j} = p_j^2$, and $g \mapsto (p_j(g))_{j \text{ odd} \leq \ell}$ induces an isomorphism $\mathcal{G}/\mathcal{G}^2 \cong \mathbb{F}_q^m$. The q^m quadratic characters are*

$$\varepsilon_\lambda(g) = (-1)^{\text{Tr}_{\mathbb{F}_q/\mathbb{F}_2} \sum_{j \text{ odd}} \lambda_j p_j(g)}, \quad \lambda \in \mathbb{F}_q^m,$$

and on a root α of a polynomial in the class g , $\varepsilon_\lambda(\alpha) = \psi_0(\sum_j \lambda_j \alpha^j)$ with ψ_0 the canonical additive character of $\mathbb{F}_{q^n}$. In particular ε_λ is primitive (does not factor through $\mathcal{G}_{\ell-1}$) if and only if $\lambda_\ell \neq 0$, which requires ℓ odd, i.e. d even.

Proof. Power sums are additive on products because roots are concatenated; $p_j(g^2) = 2p_j(g) = 0$ for odd j and $p_{2j} = p_j^2$ in characteristic 2. The map is onto $\mathbb{F}_q^m$ because $p_j = c_j + (\text{terms in } c_{<j})$ lets one solve for the odd c_j ; its kernel has order $q^{\ell-m} = |\mathcal{G}^2|$ and contains $\mathcal{G}^2$. Finally $p_j(\text{charpoly } \alpha) = \text{Tr}_{q^n/q}(\alpha^j)$ and $\text{Tr}_{q/2} \circ \text{Tr}_{q^n/q} = \text{Tr}_{q^n/2}$. $\square$

We write $\chi(x) = (-1)^{\text{Tr}_{q/2}(x)}$ for $x \in \mathbb{F}_q$ and $W = \{w \in \mathbb{F}_q : \text{Tr}_{q/2} w = 1\}$. Throughout, $\varepsilon_0 := (-1)^{k+1}$.

3. THE CASE $d = 4$: UNCONDITIONAL THEOREMS

Here $n = 8$, $\ell = 3$, $g = (c_1, c_2, c_3) = (a_7, a_6, a_5)$, $\mathcal{G}^2 = \{(0, c, 0)\}$ has order q , and $\mathcal{G}_0 := \{c_1 = 0\} \cong (\mathbb{F}_q^2, +)$ is a subgroup of order q^2 containing $\mathcal{G}^2$.

3.1. The quadratic tower.

Lemma 3.1. *For $\alpha \in \mathbb{F}_{q^8}$ put $a = \alpha + \alpha^{q^4}$, $n = \alpha^{1+q^4} \in \mathbb{F}_{q^4}$ and $\text{Tr} = \text{Tr}_{q^4/q}$. Then*

$$e_1(\alpha) = \text{Tr } a, \quad e_2(\alpha) = e_2(a) + \text{Tr } n, \quad e_3(\alpha) = e_3(a) + \text{Tr } a \text{ Tr } n + \text{Tr}(an),$$

where $e_j(a)$ are the elementary symmetric functions of the four conjugates of a . A pair (a, n) has exactly two preimages α if $t^2 + at + n$ is irreducible over $\mathbb{F}_{q^4}$ (equivalently $a \neq 0$ and $\text{Tr}_{q^4/2}(n/a^2) = 1$), one preimage if $a = 0$ ($\alpha \in \mathbb{F}_{q^4}$, $n = \alpha^2$), and none otherwise.

Proof. The eight conjugates of α are the roots of $\prod_{i < 4} (t^2 + a^{q^i} t + n^{q^i})$; expand. If $t^2 + at + n$ is reducible with $a \neq 0$ its roots $\beta \in \mathbb{F}_{q^4}$ satisfy $\beta + \beta^{q^4} = 0 \neq a$. $\square$

Two facts are used repeatedly: (i) $\text{Tr}_{q^8/2}$ vanishes on $\mathbb{F}_{q^4}$, $\text{Tr}_{q^4/q}$ vanishes on $\mathbb{F}_{q^2}$ and on $\mathbb{F}_q$; (ii) for $a \in \mathbb{F}_{q^4}$ the $\mathbb{F}_q$ -linear forms $\text{Tr}(n)$, $\text{Tr}(an)$ are independent iff $a \notin \mathbb{F}_q$, and the $\mathbb{F}_2$ -form $\text{Tr}_{q^4/2}(a^{-2}n)$ lies in the $\mathbb{F}_2$ -span of $\{\text{Tr}_{q^4/2}(cn) : c \in \mathbb{F}_q + \mathbb{F}_q a\}$ iff $a \in \mathbb{F}_{q^2}$.

3.2. The closed formula.

Theorem 3.2. *For every $q = 2^k$,*

$$\begin{aligned} \Psi(c_1, c_2, c_3) &= q^5 \quad (c_1 \neq 0), \\ \Psi(0, c_2, c_3) &= q^5 + q^3 [c_3 = 0] + 2q^3 I(c_2, c_3) - 2q^2 K(c_2, c_3), \end{aligned}$$

where $I(c_2, c_3) = [c_2 c_3 \neq 0 \text{ and } \text{Tr}_{q/2}(c_2^3/c_3^2) = 1]$ and

$$K(c_2, c_3) = \sum_{w \in W} \sum_{v \in \mathbb{F}_q} \chi \left(\frac{c_2 v^2}{w} + \frac{\sqrt{c_2} v}{w} + \frac{c_3 v^3}{w^2} \right).$$

Proof. We count pairs (a, n) as in Lemma 3.1 with $(e_1, e_2, e_3) = (c_1, c_2, c_3)$, according to the degree of a over $\mathbb{F}_q$.

Case I: $a \notin \mathbb{F}_{q^2}$. We need $\text{Tr } a = c_1$ and then two independent $\mathbb{F}_q$ -linear conditions on n , $\text{Tr } n = c_2 + e_2(a)$ and $\text{Tr}(an) = c_3 + e_3(a) + c_1(c_2 + e_2(a))$, cutting out an affine subspace of size q^2 ; the irreducibility condition is a non-constant affine $\mathbb{F}_2$ -functional on it by fact (ii), so exactly $q^2/2$ values of n qualify, each with two roots. The number of a is $q^3 - q^2[c_1 = 0]$ (all $a \in \mathbb{F}_{q^2}$ have $\text{Tr } a = 0$). Contribution: $q^5 - q^4[c_1 = 0]$.

Case II: $a \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$. Then $\text{Tr } a = 0$, so $c_1 = 0$. Let $T = a + a^q \in \mathbb{F}_q^\times$, $N = a^{1+q}$; the conjugates are a, a^q, a, a^q , so $e_2(a) = T^2$ and $e_3(a) = 0$. The conditions are $\text{Tr } n = c_2 + T^2$, $\text{Tr}(an) = c_3$ (q^2 solutions). Writing $a^{-2} = x + ya$ with $x = (N + T^2)/N^2$, $y = T/N^2$ (from $a^2 = Ta + N$), the irreducibility functional equals $\text{Tr}_{q/2}(x(c_2 + T^2) + yc_3)$ on the solution set, and $\text{Tr}_{q/2}(xT^2) = \text{Tr}_{q/2}(z + z^2) = 0$ with $z = T^2/N$. Hence the contribution is $2q^2$ times the number of a with

$$\tau(a) := \text{Tr}_{q/2} \left(\frac{(N + T^2)c_2 + Tc_3}{N^2} \right) = 1.$$

Such a correspond two-to-one to irreducible $t^2 + Tt + N$, i.e. to (T, N) with $\text{Tr}_{q/2}(N/T^2) = 1$. Substituting $N = T^2w$ ($w \in W$), $v = 1/T$ and using $\text{Tr}_{q/2}(c_2v^2/w^2) = \text{Tr}_{q/2}(\sqrt{c_2}v/w)$ gives contribution $4q^2M$ with $M = \#\{(v, w) \in \mathbb{F}_q^\times \times W : \tau = 1\} = q^2/4 - K/2$, i.e. $q^4 - 2q^2K$.

Case III: $a \in \mathbb{F}_q^\times$. Then $c_1 = 0$, $e_2(a) = e_3(a) = 0$, the conditions read $\text{Tr } n = c_2$, $a \text{Tr } n = c_3$, and irreducibility is $\text{Tr}_{q/2}(a^{-2}c_2) = 1$. This forces $c_2 \neq 0$, $a = c_3/c_2 \neq 0$ and $\text{Tr}_{q/2}(c_2^3/c_3^2) = 1$; then n ranges over q^3 elements, each with two roots. Contribution $2q^3I(c_2, c_3)$.

Case IV: $a = 0$. $\alpha \in \mathbb{F}_{q^4}$, $(e_1, e_2, e_3) = (0, (\text{Tr } \alpha)^2, 0)$; contribution $q^3[c_1 = 0][c_3 = 0]$.

Adding the four contributions gives the theorem. $\square$

Corollary 3.3. *For every $q = 2^k$:*

- (a) $\varphi = \Psi - q^5$ is supported on $\mathcal{G}_0 = \{c_1 = 0\}$.
- (b) $\Psi(0, 0, 0) = \#\{\alpha \in \mathbb{F}_{q^8} : e_1 = e_2 = e_3 = 0\} = q^5 - q^4 + q^3$; equivalently there are exactly $q^4(q-1)/8$ irreducible octics with $a_7 = a_6 = a_5 = 0$.
- (c) For $c \neq 0$, $\Psi(0, c, 0) = q^5 + \varepsilon_0 q^3$, i.e. $q^5 + q^3$ for k even and $q^5 - q^3$ for k odd.
- (d) $A := \sum_c \Psi(0, c, 0) = \#\{\alpha : \text{Tr } \alpha = \text{Tr } \alpha^3 = 0\}$ equals q^6 for k even and $q^6 - 2q^4 + 2q^3$ for k odd.

Proof. (a) is the first line of the theorem. (b): $K(0, 0) = q^2/2$, $I(0, 0) = 0$. (c): with $y = \sqrt{c}v$ one has $\chi(cv^2/w + \sqrt{c}v/w) = \chi((y^2 + y)/w)$; since $y \mapsto y^2 + y$ is two-to-one onto $\{\text{Tr } z = 0\}$ and $z \mapsto \text{Tr}(z/w)$ is nontrivial there unless $w = 1$, $K(c, 0) = q[1 \in W] = q[k \text{ odd}]$. (d) follows from (b), (c). The prime-power part of $\Psi(0, c, 0)$ equals q^3 for every c (Part I), which gives the count of irreducibles in (b). $\square$

Theorem 3.4. *For every $q = 2^k$, $\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^5} = q(q-1)$.*

Proof. By Corollary 3.3(b), (c), $\sum_{g \in \mathcal{G}^2} \varphi(g)^2 = (q^4 - q^3)^2 + (q-1)q^6 = q^7(q-1)$, and $|\mathcal{G}^2| = q$. $\square$

Remark 3.5. The factor $q(q-1) = (q-1)^2 + (q-1)$ splits as the contribution of the identity class (the interval around t^8 , whose prime deficit is $q^4 - q^3$, relative size $1/q$) plus that of the other $q-1$ Legendre classes ($\pm q^3$ each). The sign in (c) depends on the parity of k but the variance does not.

3.3. Cubic Artin–Schreier sums over $\mathbb{F}_{q^8}$. Let $S(b, a) = \sum_{x \in \mathbb{F}_{q^8}} \psi_0(ax^3 + bx)$.

Lemma 3.6. *For $a \in \mathbb{F}_q^\times$ and $b \in \mathbb{F}_q$, $S(b, a) = S(0, a)$.*

Proof. For any $c \in \mathbb{F}_{q^8}$ the substitution $x \mapsto x + c$ gives $S(b, a) = \psi_0(ac^3 + bc)S(b + T_a(c), a)$ with $T_a(c) = ac^2 + \sqrt{ac}$, since $\text{Tr}(acx^2) = \text{Tr}(\sqrt{ac}x)$. If $c \in \mathbb{F}_{q^4}$ then $\text{Tr}_{q^8/2}(ac^3 + bc) = 0$. It remains to find $c \in \mathbb{F}_{q^4}$ with $T_a(c) = b$. The map T_a preserves $\mathbb{F}_{q^4}$ and is self-adjoint for $\langle u, v \rangle = \text{Tr}_{q^4/2}(uv)$, so $\text{im } T_a = (\ker T_a)^\perp$ with $\ker T_a = \{0\} \cup \{u \in \mathbb{F}_{q^4} : u^3 = a^{-1}\}$. For such

$u \neq 0$, $\zeta = u^{q-1}$ is a cube root of unity. If $\zeta = 1$, $u \in \mathbb{F}_q$ and $\text{Tr}_{q^4/2}(bu) = \text{Tr}_{q/2}(4bu) = 0$. If $\zeta \neq 1$ and k is even, $\zeta^q = \zeta$ forces $u^{q^4} = \zeta^4 u \neq u$, impossible; if k is odd, $\zeta^q = \zeta^2$ gives $u^{q^2} = u$, so $u \in \mathbb{F}_{q^2}$ and $\text{Tr}_{q^4/2}(bu) = \text{Tr}_{q^2/2}(2bu) = 0$. Hence $b \perp \ker T_a$. $\square$

Proposition 3.7 (Carlitz's evaluation, elementary proof). *For $a \in \mathbb{F}_q^\times$ let $\nu = a$. Then*

$$S(0, a) = \sum_{x \in \mathbb{F}_{q^8}} \psi_0(ax^3) = \begin{cases} -2q^4, & a \text{ is a cube in } \mathbb{F}_{q^8}, \\ +q^4, & \text{otherwise,} \end{cases}$$

and $a \in \mathbb{F}_q^\times$ is a cube in $\mathbb{F}_{q^8}$ iff k is odd or a is a cube in $\mathbb{F}_q$.

Proof. By Lemma 3.6, $S(0, a) = q^{-1} \sum_{b \in \mathbb{F}_q} S(b, a) = \sum_{\alpha: \text{Tr } \alpha = 0} \psi_0(a\alpha^3) = \widehat{\Psi}(0, a)$ in the notation of (3) below, whose value is computed in Theorem 3.11. The cube criterion: $(q^8 - 1)/(q - 1) \equiv 8 \equiv 2 \pmod{3}$ when $q \equiv 1 \pmod{3}$, while $3 \mid (q^8 - 1)/(q - 1)$ when $q \equiv 2 \pmod{3}$. $\square$

Remark 3.8. The classical route [3] uses the cubic Gauss sum $g_4 = \sum_{y \in \mathbb{F}_4^\times} \chi_3(y)(-1)^{\text{Tr } y} = 2$ and the Hasse–Davenport relation $g_{4^m} = (-1)^{m-1} 2^m$, giving $\sum_{x \in \mathbb{F}_{2^{2m}}} \psi_0(ax^3) = (-1)^{m+1} 2^{m+1}$ for cubes and $(-1)^m 2^m$ otherwise; with $m = 4k$ this agrees with Proposition 3.7.

3.4. The full-group variance. Identify $\mathcal{G}_0 = \{(0, x, y)\}$ with $(\mathbb{F}_q^2, +)$ and set

$$(3) \quad \widehat{\Psi}(\mu, \nu) = \sum_{x, y \in \mathbb{F}_q} \Psi(0, x, y) \chi(\mu x + \nu y).$$

For $j \in \mathbb{F}_q^\times$ let $C_j(x) = x^3 + (1+j)x^2 + x + 1$, $R_j = \#\{x \in \mathbb{F}_q : C_j(x) = 0\}$, $L_j = \#\{x : C_j(x) = 0, \text{Tr}_{q/2} x = 1\}$ and $L'_j = R_j - L_j$.

Lemma 3.9. *For $(\mu, \nu) \neq (0, 0)$:*

$$\begin{aligned} \widehat{\Psi}(\mu, 0) &= -q^4 \quad (\mu \neq 0), & \widehat{\Psi}(0, \nu) &= q^4(1 - R_0(\nu) - 2[k \text{ odd}]) \quad (\nu \neq 0), \\ \widehat{\Psi}(\mu, \nu) &= q^4(1 - 3L_j - L'_j) & & (\mu\nu \neq 0, j = \mu^3/\nu^2), \end{aligned}$$

where $R_0(\nu) = \#\{t : t^3 = \nu\}$ equals 1 for k odd and $3[\nu \text{ is a cube}]$ for k even. In particular $\widehat{\Psi}(0, \nu) = -2q^4$ for k odd, and for k even $-2q^4$ or $+q^4$ according as ν is a cube in $\mathbb{F}_q$ or not.

Proof. Transform the three terms of Theorem 3.2. The term $q^3[y = 0]$ gives $q^4[\mu = 0]$. For K , the identity $\sum_x \chi(x\alpha + \sqrt{x}\beta) = q[\sqrt{\alpha} = \beta]$ yields $\widehat{K}(\mu, \nu) = q^2 \#\{(v, w) \in \mathbb{F}_q \times W : v^3 = \nu w^2, v^2(w+1) = \mu w^2\}$. For $I = [x, y \neq 0] \frac{1-\chi(x^3/y^2)}{2}$, the substitution $x = yt$ gives $\sum_{x, y \neq 0} \chi(x^3/y^2 + \mu x + \nu y) = q R^*(\mu, \nu) - (q-1)$ with $R^* = \#\{t \neq 0 : t^3 + \mu t + \nu = 0\}$. The three cases follow; for $\mu\nu \neq 0$, dividing the two conditions in $\widehat{K}$ gives $v = \nu(w+1)/\mu$ and $\nu^2(w+1)^3 = \mu^3 w^2$, and $t = (\nu/\mu)s$, $s = 1 + 1/w$ transforms $t^3 + \mu t + \nu = 0$ into $C_j(w) = 0$, so $R^* = R_j$ and $\widehat{K} = q^2 L_j$. $\square$

Lemma 3.10. $\Sigma := \sum_{j \in \mathbb{F}_q^\times} (1 - 3L_j - L'_j)^2 = 3q - 4 + (-1)^k$.

Proof. Put $c(x) = 2 - \chi(x) \in \{1, 3\}$, so $3L_j + L'_j = \sum_{C_j(x)=0} c(x)$. Every $x \in \mathbb{F}_q \setminus \{0, 1\}$ is a root of exactly one C_j ($j = (x+1)^3/x^2$), and 0, 1 are roots of none. Expanding, $\Sigma = (q-1) - 2 \sum_{x \neq 0, 1} c(x) + \sum_{x \neq 0, 1} c(x)^2 + \text{OFF}$, where OFF is the sum of $c(x)c(x')$ over ordered pairs of *distinct* roots of a common C_j . Using $\sum_{x \neq 0, 1} \chi(x) = -1 - (-1)^k$ one finds $\Sigma = 2q - 1 + 2(-1)^k + \text{OFF}$. Distinct roots $x \neq x'$ of a common C_j are exactly the pairs with $x, x' \notin \{0, 1\}$ satisfying

$$(4) \quad x^2 x'^2 + x x' + x + x' = 0$$

(the third root is $1/(xx')$ and $e_2 = 1$; conversely (4) with $x, x' \notin \{0, 1\}$ produces a cubic C_j with $j \neq 0$). On (4) one has $x + x' = p^2 + p$ with $p = xx'$, hence $\chi(x)\chi(x') = 1$, and by symmetry $\text{OFF} = 5P - 4 \sum_{\text{pts}} \chi(x)$ where P is the number of points. For fixed $x \notin \{0, 1\}$, (4) is a quadratic

in x' with $1 + \chi(x^3/(x+1)^2)$ roots, all $\notin \{0, 1\}$. With $y = x+1$: $x^3/(x+1)^2 = y+1+1/y+1/y^2$, whose χ -value is $(-1)^k \chi(y)$, and $x + x^3/(x+1)^2 = 1/y + 1/y^2$, whose χ -value is 1. Therefore $P = \sum_{\text{pts}} \chi(x) = q - 3 - (-1)^k$, $\text{OFF} = q - 3 - (-1)^k$ and $\Sigma = 3q - 4 + (-1)^k$. $\square$

Theorem 3.11. *For every $q = 2^k$, $\frac{\text{Var}_{\mathcal{G}}(\Psi)}{q^5} = 3 - \frac{3}{q}$ and $\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{\text{Var}_{\mathcal{G}}(\Psi)} = \frac{q^2}{3}$.*

Proof. By Corollary 3.3(a) and Parseval on $\mathcal{G}_0 \cong \mathbb{F}_q^2$ (note $\sum_{\mathcal{G}_0} \varphi = 0$),

$$\sum_{\mathcal{G}} \varphi^2 = \frac{1}{q^2} \sum_{(\mu, \nu) \neq (0,0)} |\widehat{\Psi}(\mu, \nu)|^2.$$

Lemma 3.9 gives $(q-1)q^8$ from $\nu = 0$, $\kappa_k(q-1)q^8$ from $\mu = 0$ with $\kappa_k = 4$ (k odd) and $\kappa_k = \frac{1}{3} \cdot 4 + \frac{2}{3} \cdot 1 = 2$ (k even), and $q^8(q-1)\Sigma$ from $\mu\nu \neq 0$ (each j has exactly $q-1$ preimages (μ, ν)). By Lemma 3.10 the bracket $1 + \kappa_k + \Sigma$ equals $3q$ in both parities, so $\sum_{\mathcal{G}} \varphi^2 = 3q^7(q-1)$. $\square$

Theorem 3.12. *Let $\varepsilon_{\lambda_1, \lambda_3}$ be the quadratic characters of $\mathcal{G}_3$ (Proposition 2.2). Then*

- (i) $\rho(\varepsilon_{\lambda_1, 0}) = 1$ for all $\lambda_1 \neq 0$ ($q-1$ characters of conductor u^2);
- (ii) $\rho(\varepsilon_{\lambda_1, \lambda_3}) = r(\lambda_3)$ is independent of λ_1 and constant on cosets of the cubes in $\mathbb{F}_q^\times$;
- (iii) if k is odd, $r \equiv \frac{q-3}{3(q-1)}$;
- (iv) $\rho_{\text{avg}} = \frac{q^2/3-1}{q^2-1} \rightarrow \frac{1}{3}$.

Proof. (i),(ii): φ is supported on $\mathcal{G}_0$ where $p_1 = 0$ and $p_3 = c_3$, so $\varepsilon_{\lambda_1, \lambda_3}|_{\mathcal{G}_0} = \chi(\lambda_3 c_3)$; the scaling automorphism $(c_1, c_2, c_3) \mapsto (sc_1, s^2 c_2, s^3 c_3)$ preserves Ψ . (iii): for k odd cubing is onto, so $\Phi(y) = \sum_x \varphi(0, x, y)^2$ is constant for $y \neq 0$, equal to $(3q^7(q-1) - q^7(q-1))/(q-1) = 2q^7$ by Theorems 3.4–3.11; then $r = (q^7(q-1) - 2q^7)/(3q^7(q-1))$. (iv) is immediate. $\square$

Remark 3.13. For $q = 8$ this gives seven characters with $\rho = 1$ and 56 with $\rho = 5/21 = 0.2381$, matching [4, Table 4] exactly; the “order-two coordinates” there contribute $7 \cdot 5/21 = 5/3$. The excess over Keating–Rudnick comes from the primitive quadratic characters, whose ρ does not decay in q ; $\sum_{\chi \neq 1} \psi_8(\chi) = -q^7 + q^6$ shows that $\mathbb{E}_\chi \text{Tr } \Theta_\chi^8 \rightarrow -1$, so the Frobenius classes are not Haar-distributed in $U(2)$ when $p = 2$, $\ell = 3$.

4. THE CASE $d = 6$: THE THREE-VALUE STRUCTURE THEOREM

Now $n = 12$, $\ell = 5$, $\mathcal{G}^2 = \{(0, c_2, 0, c_4, 0)\}$ has order q^2 and

$$U := \{\alpha \in \mathbb{F}_{q^{12}} : \text{Tr } \alpha = \text{Tr } \alpha^3 = \text{Tr } \alpha^5 = 0\}, \quad \text{Tr} = \text{Tr}_{q^{12}/q}.$$

Since $p_3 = c_3 + c_1 c_2 + c_1^3$ and $p_5 \equiv c_5$ modulo terms involving c_1, c_3 , the Legendre classes are exactly the classes with $p_1 = p_3 = p_5 = 0$, and $\Psi(0, c_2, 0, c_4, 0) = \#\{\alpha \in U : e_2(\alpha) = c_2, e_4(\alpha) = c_4\}$. Define

$$A_6 = |U|, \quad X = \sum_{\alpha \in U} \chi(e_2(\alpha)), \quad Y = \sum_{\alpha \in U} \chi(e_4(\alpha)).$$

Lemma 4.1. *U is stable under $\alpha \mapsto \alpha + c$ ($c \in \mathbb{F}_q$) and $\alpha \mapsto s\alpha$ ($s \in \mathbb{F}_q^\times$). On U ,*

$$e_2(\alpha + c) = e_2(\alpha), \quad e_4(\alpha + c) = e_4(\alpha) + c^2 e_2(\alpha) + c^4, \quad (e_2, e_4)(s\alpha) = (s^2 e_2, s^4 e_4)(\alpha).$$

Proof. $\text{Tr}(\alpha + c)^3 = \text{Tr } \alpha^3 + c(\text{Tr } \alpha)^2$ and $\text{Tr}(\alpha + c)^5 = \text{Tr } \alpha^5 + c(\text{Tr } \alpha)^4$ since $12c = 0$. For the elementary symmetric functions, $e_j(\alpha + c) = \sum_i \binom{12-i}{j-i} c^{j-i} e_i(\alpha)$; modulo 2, $\binom{12}{2} = 66$, $\binom{11}{1} = 11$, $\binom{12}{4} = 495$, $\binom{11}{3} = 165$, $\binom{10}{2} = 45$, $\binom{9}{1} = 9$, and on U one has $e_1 = e_3 = 0$. $\square$

Theorem 4.2 (three-value profile). *For all $q = 2^k$ and all $(c_2, c_4) \in \mathbb{F}_q^2$,*

$$\Psi(0, c_2, 0, c_4, 0) = q^7 + \frac{A_6 - q^9 - X}{q^2} + \frac{X}{q} [c_2 = 0] + \frac{Y}{q} [c_2 \neq 0] \chi\left(\frac{c_4}{c_2^2}\right).$$

Equivalently, the Fourier transform $\widehat{\Psi}_H(\mu, \nu) = \sum_{\alpha \in U} \chi(\mu e_2 + \nu e_4)$ equals $A_6, X, Y\chi(\mu/\sqrt{\nu})$ for $(\mu, \nu) = (0, 0), (\mu \neq 0, 0), \nu \neq 0$ respectively.

Proof. Let $F(\lambda) = \sum_U \chi(\lambda e_2 + e_4)$. By Lemma 4.1, $F(\lambda) = \sum_U \chi(\lambda e_2 + e_4 + c^2 e_2 + c^4) = \chi(c)F(\lambda + c^2)$, hence $F(\lambda) = \chi(\sqrt{\lambda})F(0) = \chi(\lambda)Y$. For $\nu \neq 0$ choose $s = \nu^{-1/4}$; scaling gives $\widehat{\Psi}_H(\mu, \nu) = F(\mu\nu^{-1/2}) = \chi(\mu/\sqrt{\nu})Y$, and for $\mu \neq 0$ scaling gives $\widehat{\Psi}_H(\mu, 0) = X$. Fourier inversion on $\mathbb{F}_q^2$ yields the displayed formula. $\square$

Corollary 4.3. Put $B = (A_6 - q^9 - X)/q^2$. Then $\varphi = \Psi - q^7$ equals $\varphi_0 = B + X/q$ on the q classes with $c_2 = 0$, $\varphi_A = B + Y/q$ on the $q(q-1)/2$ classes with $c_2 \neq 0$, $\text{Tr}_{q/2}(c_4/c_2^2) = 0$, and $\varphi_B = B - Y/q$ on the remaining $q(q-1)/2$ classes, and

$$(5) \quad \frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \frac{q\varphi_0^2 + q(q-1)(B^2 + Y^2/q^2)}{q^9}.$$

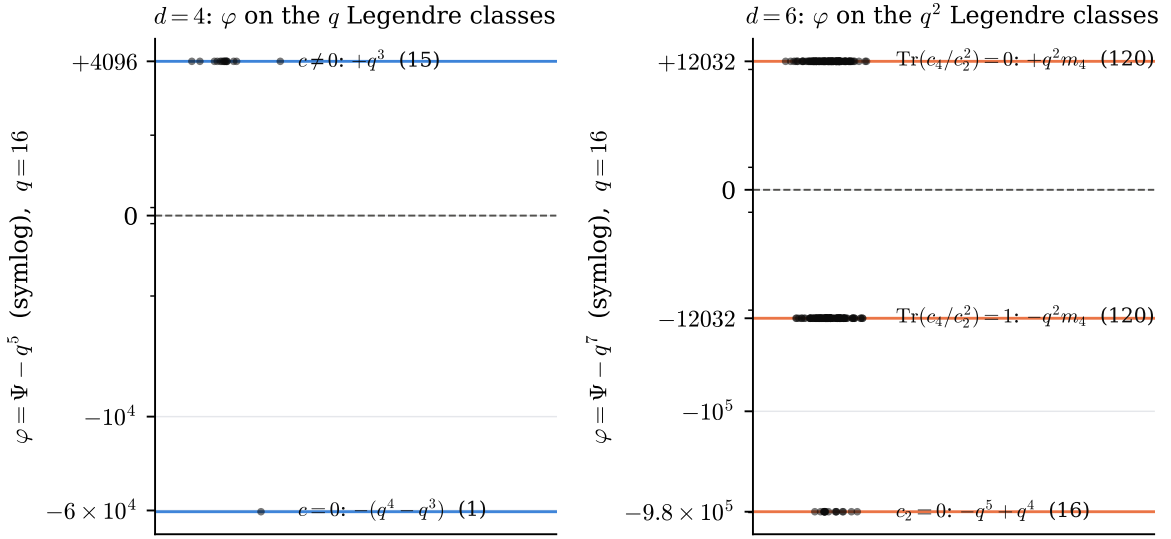


FIGURE 1. The Legendre deviations $\varphi = \Psi - q^{d+1}$ for $q = 16$. Left: $d = 4$, the q classes $(0, c, 0)$ take the two values of Corollary 3.3; the identity class carries $-(q^4 - q^3)$. Right: $d = 6$, the q^2 classes $(0, c_2, 0, c_4, 0)$ take exactly the three values of Theorem 4.2 ($m_4 = 47$). Each dot is one class (horizontal jitter only); the deviations are not continuous noise but are pinned to discrete levels by the translation symmetry of U .

Remark 4.4. The same argument applies to every d : $U_d = \{p_j = 0, j \text{ odd} \leq \ell\}$ is stable under $\mathbb{F}_q$ -translations and scalings, and the transformation law of e_{2j} is governed by $\binom{2d-i}{2j-i} \bmod 2$ (Lucas' theorem). For $d = 4$ one recovers Corollary 3.3(c).

5. THE QUADRATIC TOWER $\mathbb{F}_{q^{12}} \supset \mathbb{F}_{q^6}$

Lemma 5.1. For $\alpha \in \mathbb{F}_{q^{12}}$ let $a = \alpha + \alpha^{q^6}$, $n = \alpha^{1+q^6} \in \mathbb{F}_{q^6}$, $\text{Tr} = \text{Tr}_{q^6/q}$, and $e_j(a)$ the elementary symmetric functions of the six conjugates of a . Then

$$\begin{aligned} e_1 &= \text{Tr} a, & e_2 &= e_2(a) + \text{Tr} n, & e_3 &= e_3(a) + \text{Tr} a \text{Tr} n + \text{Tr}(an), \\ e_4 &= e_4(a) + e_2(a) \text{Tr} n + e_1(a) \text{Tr}(an) + \text{Tr}(a^2 n) + e_2(n), \\ e_5 &= e_5(a) + e_3(a) \text{Tr} n + e_2(a) \text{Tr}(an) + e_1(a) \text{Tr}(a^2 n) + \text{Tr}(a^3 n) \\ &\quad + e_1(a) e_2(n) + \text{Tr}(an) \text{Tr} n + \text{Tr}(an^2), \\ p_3 &= \text{Tr}(a^3 + an), & p_5 &= \text{Tr}(a^5 + a^3 n + an^2), \end{aligned}$$

and $e_2(\beta + \beta^q) = \text{Tr}_{q^{12}/q}(\beta^{q+1}) + (\text{Tr}_{q^{12}/q} \beta)^2$ for all $\beta \in \mathbb{F}_{q^{12}}$. The multiplicity statement of Lemma 3.1 holds verbatim with $\mathbb{F}_{q^4}$ replaced by $\mathbb{F}_{q^6}$; in particular $\mathbb{F}_{q^6} \subset U$, with $e_2|_{\mathbb{F}_{q^6}} = (\text{Tr } \alpha)^2$ and $e_4|_{\mathbb{F}_{q^6}} = e_2(\alpha)^2$.

Proof. Expand $\prod_{i < 6} (t^2 + a_i t + n_i)$ using $e_2(a \setminus j) = e_2(a) + a_j e_1(a) + a_j^2$, $e_3(a \setminus j) = e_3(a) + a_j e_2(a) + a_j^2 e_1(a) + a_j^3$ and $e_1(a \setminus j, k) = e_1(a) + a_j + a_k$. The power sums follow from $s_k = a s_{k-1} + n s_{k-2}$ for $s_k = \alpha^k + \alpha^{q^6 k}$. All identities were also checked by exhaustive computation over $\mathbb{F}_{2^{12}}$. $\square$

Lemma 5.2 (Gauss sum of e_2). *On $\mathbb{F}_{q^6}$ the quadratic form e_2 has polar form $B(x, y) = \text{Tr } x \text{Tr } y + \text{Tr}(xy)$, which is nondegenerate. For every $\beta \in \mathbb{F}_{q^6}$, $\sum_{n \in \mathbb{F}_{q^6}} \chi(e_2(n) + \text{Tr}(\beta n)) = \chi(e_2(\beta)) G$ with $G = \sum_n \chi(e_2(n))$.*

Proof. $n_0 = \beta + \text{Tr } \beta$ satisfies $B(n_0, \cdot) = \text{Tr}(\beta \cdot)$, and $e_2(\beta + c) = e_2(\beta) + c \text{Tr } \beta + c^2$ for $c \in \mathbb{F}_q$ gives $e_2(n_0) = e_2(\beta)$. $\square$

Theorem 5.3 (tower reduction). *Let a range over $\mathbb{F}_{q^6} \setminus \{0\}$ with $\text{Tr } a = 0$, $(\lambda, \mu) \in \mathbb{F}_q^2$, $\varepsilon \in \{0, 1\}$, and put*

$$\beta' = \lambda a + \sqrt{\mu} \sqrt{a} + \mu a^3 + \varepsilon a^{-2}, \quad \beta = e_2(a) + a^2 + \beta', \quad w = \chi(\lambda \text{Tr } a^3 + \mu \text{Tr } a^5).$$

Then

$$\begin{aligned} A_6 &= q^6 + q^4 \sum_a \sum_{\beta'=0} (-1)^\varepsilon w, & X &= q^4 \sum_a \chi(e_2(a)) \sum_{\beta'=1} (-1)^\varepsilon w, \\ Y &= G \left[1 + q^{-2} \sum_a \sum_{\lambda, \mu, \varepsilon} (-1)^\varepsilon \chi(e_4(a) + \lambda \text{Tr } a^3 + \mu \text{Tr } a^5 + e_2(\beta)) \right]. \end{aligned}$$

Proof. For $a \neq 0$ with $\text{Tr } a = 0$, the conditions $p_3 = p_5 = 0$ read $\text{Tr}(an) = \text{Tr } a^3$ and $\text{Tr}(\sqrt{a}n)^2 + \text{Tr}(a^3 n) = \text{Tr } a^5$ (using $\text{Tr}(an^2) = \text{Tr}(\sqrt{a}n)^2$); expand their indicators and the irreducibility indicator $\frac{1}{2} \sum_\varepsilon (-1)^\varepsilon \chi(\varepsilon \text{Tr}(a^{-2}n))$ by additive characters, using $\chi(\mu t^2) = \chi(\sqrt{\mu} t)$. The n -sums are $q^6[\gamma = 0]$ for A_6, X (no $e_2(n)$ term) and $\chi(e_2(\beta))G$ for Y by Lemma 5.2. The contribution of $\alpha \in \mathbb{F}_{q^6}$ is $q^6, 0, G$ respectively. $\square$

Remark 5.4. The equation $\beta' = 0$ with $\varepsilon = 0$ and $\mu \neq 0$ squares to $\mu^2 a^5 + \lambda^2 a + \mu = 0$, forcing $a \in \mathbb{F}_{q^2} \cup \mathbb{F}_{q^3}$ (a thin layer, as in Case II of Theorem 3.2). For $\varepsilon = 1$, with $u = \mu a^5 + \lambda a^3 + 1$ it becomes $u^2 + u = \lambda a^3 + 1$: the points $(a, u) \in E_\lambda(\mathbb{F}_{q^6})$ of the supersingular elliptic curve $E_\lambda : y^2 + y = \lambda x^3 + 1$ with $(u + \lambda a^3 + 1)/a^5 \in \mathbb{F}_q$. The additional term $e_2(\beta)$ in Y is quadratic in $(\lambda, \sqrt{\mu})$ with cubic cross terms, which is why Y does not collapse onto a thin layer.

The reductions were verified numerically for $q = 2, 4, 8$; they give $G = \varepsilon_0 q^3$ for $k \leq 3$.

6. QUADRATIC FORMS AND SUPERSINGULAR ARTIN-SCHREIER CURVES

6.1. A_6 and X as sums of Frobenius traces.

Theorem 6.1. *Write $\lambda = (\lambda_1, \lambda_3, \lambda_5) \in \mathbb{F}_q^3$ and $S(\lambda) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(\lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x)$. Then*

$$(6) \quad A_6 - q^9 = \frac{1}{q^3} \sum_{\lambda \neq 0} S(\lambda),$$

and $S(\lambda) = -\sum_j \omega_j(\lambda)^{12}$, where $\omega_j(\lambda)$ are the Frobenius eigenvalues over $\mathbb{F}_q$ of the Artin-Schreier curve $C_\lambda : y^2 + y = \lambda_5 x^5 + \lambda_3 x^3 + \lambda_1 x$ (genus 2, 1, 0 according as $\lambda_5 \neq 0$; $\lambda_5 = 0 \neq \lambda_3$; only $\lambda_1 \neq 0$). Moreover $\text{Tr}_{q^{12}/2}(\lambda_5 x^5 + \lambda_3 x^3) = Q_\lambda(x)$ is an $\mathbb{F}_2$ -quadratic form with polar form $B_\lambda(x, y) = \text{Tr}(y T_\lambda(x))$,

$$T_\lambda(x) = \lambda_5 x^4 + (\lambda_5 x)^{1/4} + \lambda_3 x^2 + (\lambda_3 x)^{1/2}, \quad R_\lambda = \ker T_\lambda = \{x : \lambda_5^4 x^{16} + \lambda_3^4 x^8 + \lambda_3^2 x^2 + \lambda_5 x = 0\},$$

so that $S(\lambda) \in \{0, \pm 2^{(12k+r_\lambda)/2}\}$ with $r_\lambda = \dim_{\mathbb{F}_2} R_\lambda \leq 4$, $S(\lambda) = 0$ iff $\text{Tr}(\lambda_1 x) \neq Q_\lambda(x)$ for some $x \in R_\lambda$, and the sign otherwise is the Arf invariant of Q_λ on $\mathbb{F}_{q^{12}}/R_\lambda$ (shifted by $Q_\lambda(x_0)$, $B_\lambda(x_0, \cdot) = \text{Tr}(\lambda_1 \cdot)$). In particular all C_λ are supersingular: $\omega_j(\lambda) = \sqrt{q} \zeta_j$ with roots of unity ζ_j .

Proof. $A_6 = \sum_{g \in \mathcal{G}^2} \Psi(g) = \frac{|\mathcal{G}^2|}{|\mathcal{G}|} \sum_{\varepsilon^2=1} \psi_{12}(\varepsilon)$ and $\psi_{12}(\varepsilon_\lambda) = S(\lambda)$ by Proposition 2.2; $S(0) = q^{12}$. The L -function interpretation is standard. $Q_\lambda(x) = \text{Tr}(x(\lambda_5 x^4 + \lambda_3 x^2))$ is of the form $\text{Tr}(xL(x))$ with L linearised, whence the quadratic-form statements; the kernel of T_λ equals that of T_λ^4 . Since the same structure persists over every $\mathbb{F}_{q^m}$, $\sum_{x \in \mathbb{F}_{q^m}} (-1)^{\text{Tr} f_\lambda(x)} \in \{0, \pm 2^{(mk+r)/2}\}$ for all m , which forces the normalised eigenvalues to be roots of unity [8]. $\square$

Summing over λ_1 and using Hilbert 90 ($\alpha = \beta + \beta^q$, $\theta = \sigma + \sigma^{-1}$, $\text{Tr} \alpha^3 = \text{Tr}(\beta^2 \theta \beta)$, $\text{Tr} \alpha^5 = \text{Tr}(\beta^4 \theta \beta)$) one obtains the equivalent forms

$$(7) \quad A_6 = q^9 + \frac{1}{q^3} \sum_{(\lambda_3, \lambda_5) \neq 0} W(\lambda_3, \lambda_5), \quad X = \frac{1}{q^3} \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5),$$

with $W = \sum_{\beta} (-1)^{\text{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta)}$ and $W' = \sum_{\beta} (-1)^{\text{Tr}(\lambda_3 \beta^2 \theta \beta + \lambda_5 \beta^4 \theta \beta + \beta^{q+1} + \beta)}$ (the second uses Lemma 5.1). Their polar forms are $\text{Tr}(\gamma M(\theta \beta))$ with $M(y) = \lambda_3 y^2 + \sqrt{\lambda_3 y} + \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ ($+y$ for W'), so the radicals are $\theta^{-1}(\ker M \cap \text{im } \theta)$, of dimension $2k + \dim(\ker M \cap \text{im } \theta)$, where $\ker \theta = \mathbb{F}_{q^2}$ and $\text{im } \theta = \ker \text{Tr}_{q^{12}/q^2}$.

6.2. The pure cubic part.

Lemma 6.2. *For all k and all $\lambda_3 \in \mathbb{F}_q^\times$: $W(\lambda_3, 0) = 0$ if k is odd and $W(\lambda_3, 0) = -2q^7$ if k is even.*

Proof. $\ker M = \{0\} \cup \{y : y^3 = \lambda_3^{-1}\}$, and all three cube roots lie in $\mathbb{F}_{q^{12}}$. With $\zeta = y^{q-1} \in \mu_3$: $\zeta = 1$ gives $y \in \mathbb{F}_q$; for $\zeta \neq 1$, k odd gives $y \in \mathbb{F}_{q^2}$ and k even gives $y \in \mathbb{F}_{q^3}$. In all cases $\text{Tr}_{q^{12}/q^2} y = 0$, so $\ker M \subset \text{im } \theta$ and $\dim R = 2k + 2$, $|W| \in \{0, 2^{7k+1}\}$. Now $Q(\beta) = \text{Tr}_{q^{12}/2}(\lambda_3 \beta^2 \theta \beta)$ vanishes on $\ker \theta$, and for $\theta \beta = y$ it is independent of the choice of β (the ambiguity $c \in \mathbb{F}_{q^2}$ contributes $\text{Tr}(\lambda_3 c^2 y)$ with $c^2 y$ in a subfield of even index). If $y \in \mathbb{F}_{q^2}$, $\zeta \neq 1$: on $\mathbb{F}_{q^4}$ one has $\theta \beta = \sigma(\text{Tr}_{q^4/q^2} \beta)$, so pick $\beta_y \in \mathbb{F}_{q^4}$ with $\text{Tr}_{q^4/q^2} \beta_y = y^q$; then $Q(\beta_y) = \text{Tr}_{q^4/2}(\lambda_3 y \beta_y^2) = \text{Tr}_{q^2/2}(\lambda_3 y (\text{Tr}_{q^4/q^2} \beta_y)^2) = \text{Tr}_{q^2/2}(\lambda_3 y^{2q+1}) = \text{Tr}_{q^2/2}(\zeta^2) = k \pmod{2}$, which is 1 for k odd: $Q|_R \neq 0$ and $W = 0$. If $y \in \mathbb{F}_q$ the same computation gives $\text{Tr}_{q^2/2}(1) = 0$; if $y \in \mathbb{F}_{q^3}$ (k even) then $\text{Tr}_{q^3/q} y = 0$, θ is invertible on the trace-zero part of $\mathbb{F}_{q^3}$, $\beta_y \in \mathbb{F}_{q^3}$ and $Q(\beta_y) = 0$. So for k even $|W| = 2q^7$. Finally $W = \sum_{\lambda_1 \in \mathbb{F}_q} S(\lambda_1, \lambda_3, 0)$ with $|S| = 2q^6$ for each term ($r = 2$), and $|W| = q \cdot 2q^6$ forces all q signs to agree with that of $S(0, \lambda_3, 0) = -2q^6$ (Hasse–Davenport, $m = 6k$). $\square$

Remark 6.3. For k odd the analogue of Lemma 3.6 fails over $\mathbb{F}_{q^{12}}$: the cube roots lie in $\mathbb{F}_{q^2}$ and $[\mathbb{F}_{q^6} : \mathbb{F}_{q^2}] = 3$ is odd, so $\mathbb{F}_q \not\subset T(\mathbb{F}_{q^6})$. The vanishing of $W(\lambda_3, 0)$ is exactly this failure.

6.3. The pure quintic part. The quintic analogue uses a pure Gauss sum of order 5. Since $2^2 \equiv -1 \pmod{5}$, Stickelberger’s theorem on pure Gauss sums (see [1, Thm. 11.6.3]) gives, for every character χ of order 5 of $\mathbb{F}_{2^{4s}}^\times$, $G(\chi) = \sum_y \chi(y) (-1)^{\text{Tr} y} = (-1)^{s-1} 2^{2s}$. With $4s = 12k$, i.e. $s = 3k$, and $a \in \mathbb{F}_q^\times$,

$$(8) \quad S(0, 0, a) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(ax^5) = \sum_{j=1}^4 \chi^{-j}(a) G(\chi^j) = \varepsilon_0 q^6 (5[a \in (\mathbb{F}_{q^{12}}^\times)^5] - 1),$$

and $a \in \mathbb{F}_q^\times$ is a fifth power in $\mathbb{F}_{q^{12}}$ iff $4 \nmid k$ or $a \in (\mathbb{F}_q^\times)^5$ (because $(q^{12} - 1)/(q - 1) \equiv 0 \pmod{5}$) iff $q \not\equiv 1 \pmod{5}$ iff $4 \nmid k$.

Lemma 6.4 (pure quintic part). *For all k and all $\lambda_5 \in \mathbb{F}_q^\times$:*

$$W(0, \lambda_5) = \begin{cases} 2q^7, & k \text{ odd}, \\ 0, & k \equiv 2 \pmod{4}, \\ -4q^7, & k \equiv 0 \pmod{4} \text{ and } \lambda_5 \in (\mathbb{F}_q^\times)^5, \\ +q^7, & k \equiv 0 \pmod{4} \text{ and } \lambda_5 \notin (\mathbb{F}_q^\times)^5. \end{cases}$$

Proof. Here $M(y) = \lambda_5 y^4 + (\lambda_5 y)^{1/4}$ and $\ker M = \{0\} \cup \{y : y^{15} = \lambda_5^{-3}\}$. The equation $y^{15} = \lambda_5^{-3}$ is solvable in $\mathbb{F}_{q^{12}}$ iff λ_5 is a fifth power there (if $y^{15} = w^3$ then $y^5 = w\omega$ with $\omega^3 = 1$, and $\omega \in (\mathbb{F}_{q^{12}}^\times)^5$), and then, with $y_0^5 = \lambda_5^{-1}$, $\ker M = y_0 \mathbb{F}_{16}$ since $\mu_{15} \cup \{0\} = \mathbb{F}_{16} \subset \mathbb{F}_{q^{12}}$.

Case 4 $k \mid \lambda_5 \notin (\mathbb{F}_q^\times)^5$. $\ker M = 0$, the radical of the β -form is $\ker \theta = \mathbb{F}_{q^2}$ of dimension $2k$, Q vanishes on it, so $|W| = 2^{(12k+2k)/2} = q^7$. Each $S(\lambda_1, 0, \lambda_5)$ has trivial radical, hence $|S| = q^6$ for all q values of λ_1 ; $|W| = q \cdot q^6$ forces all signs to agree with that of $S(0, 0, \lambda_5) = -\varepsilon_0 q^6 = +q^6$ by (8). So $W = q^7$.

Case $\lambda_5 \in (\mathbb{F}_{q^{12}}^\times)^5$, k even. If $4 \mid k$ choose $y_0 \in \mathbb{F}_q$; if $k \equiv 2 \pmod{4}$ then $5 \mid q+1$, so every element of $\mathbb{F}_q$ is a fifth power in $\mathbb{F}_{q^2}$ and we choose $y_0 \in \mathbb{F}_{q^2}$. Also $\mathbb{F}_{16} \subset \mathbb{F}_{q^2}$. Hence $\ker M = y_0 \mathbb{F}_{16} \subset \mathbb{F}_{q^2} \subset \text{im } \theta$ and the radical $R = \theta^{-1}(y_0 \mathbb{F}_{16})$ has dimension $2k+4$. For $y \in y_0 \mathbb{F}_{16}$ pick $\beta_y \in \mathbb{F}_{q^4}$ with $\text{Tr}_{q^4/q^2} \beta_y = y^q$ (as in Lemma 6.2); then $Q(\beta_y) = \text{Tr}_{q^2/2}(\lambda_5 y^{4q+1})$. Write $y = y_0 c$, $c \in \mathbb{F}_{16}$, and $\rho = y_0^{q-1} \in \mu_5$ ($\rho = 1$ if $4 \mid k$). Then $\lambda_5 y^{4q+1} = \rho^4 c^{4q+1}$, and $c^{4q+1} = c^5$ if $q \equiv 1 \pmod{15}$ ($4 \mid k$), $= c^2$ if $q \equiv 4 \pmod{15}$ ($k \equiv 2 \pmod{4}$). Since $c \in \mathbb{F}_{16} \subset \mathbb{F}_{q^2}$, $\text{Tr}_{q^2/2}(x) = \frac{k}{2} \text{Tr}_{16/2}(x)$ for $x \in \mathbb{F}_{16}$. If $4 \mid k$ this is 0: $Q|_R = 0$, $|W| = 2^{(14k+4)/2} = 4q^7$; on the x -side the radical $y_0 \mathbb{F}_{16} \subset \mathbb{F}_q$ carries $Q_x(y_0 c) = \text{Tr}_{q^{12}/2}(c^5) = 3k \text{Tr}_{16/2}(c^5) = 0$ ($c^5 \in \mathbb{F}_4$) and the linear forms $\text{Tr}(\lambda_1 y_0 c) = \text{Tr}_{q/2}(12\lambda_1 y_0 c) = 0$, so all q sums $S(\lambda_1, 0, \lambda_5)$ have modulus $2^{(12k+4)/2} = 4q^6$ and $|W| = q \cdot 4q^6$ forces equal signs, that of $S(0, 0, \lambda_5) = 4\varepsilon_0 q^6 = -4q^6$: $W = -4q^7$. If $k \equiv 2 \pmod{4}$ then $Q(\beta_{y_0 c}) = \text{Tr}_{16/2}(\rho^4 c^2) = \text{Tr}_{16/2}(\rho^2 c)$, which is nonzero for suitable c ; so $Q|_R \neq 0$ and $W = 0$.

Case k odd. Fifth powers are bijective on $\mathbb{F}_q^\times$, so $y_0 \in \mathbb{F}_q$. Now $\mathbb{F}_{16} \not\subset \mathbb{F}_{q^2}$ and $q^2 \equiv 4 \pmod{15}$, so for $c \in \mathbb{F}_{16}$, $\text{Tr}_{q^{12}/q^2}(y_0 c) = y_0 \sum_{i < 6} c^{4^i} = y_0(c + c^4)$, which vanishes iff $c \in \mathbb{F}_4$. Thus $\ker M \cap \text{im } \theta = y_0 \mathbb{F}_4$ and $\dim R = 2k+2$. For $y = y_0 c$, $c \in \mathbb{F}_4^\times$: $Q(\beta_y) = \text{Tr}_{q^2/2}(\lambda_5 y_0^5 c^{4q+1}) = \text{Tr}_{q^2/2}(1) = 0$ because $4q+1 \equiv q+1 \equiv 0 \pmod{3}$. Hence $|W| = 2^{(14k+2)/2} = 2q^7$. On the x -side, $Q_x(y_0 c) = \text{Tr}_{q^{12}/2}(c^5) = 0$ as before, while $\text{Tr}_{q^{12}/2}(\lambda_1 y_0 c) = \text{Tr}_{q/2}(\lambda_1 y_0) \text{Tr}_{16/2}(c)$ (the twelve conjugates of c under q -Frobenius run three times through c, c^2, c^4, c^8); so exactly the $q/2$ values of λ_1 with $\text{Tr}_{q/2}(\lambda_1 y_0) = 0$ give $|S| = 4q^6$, the others give 0, and $|W| = \frac{q}{2} \cdot 4q^6$ forces equal signs, that of $S(0, 0, \lambda_5) = 4\varepsilon_0 q^6 = +4q^6$. So $W = 2q^7$. $\square$

Lemmas 6.2 and 6.4 give, for the pure terms, $\sum_{\lambda_3 \neq 0} W(\lambda_3, 0) + \sum_{\lambda_5 \neq 0} W(0, \lambda_5) = 2(q-1)q^7$ for k odd and $= -2(q-1)q^7$ for k even (for $4 \mid k$ the quintic terms contribute $\frac{q-1}{5}(-4q^7) + \frac{4(q-1)}{5}q^7 = 0$). Hence Conjecture 6.5 for A_6 is equivalent to

$$(9) \quad \sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = \begin{cases} (q-1)(q-2)q^7, & k \text{ odd}, \\ -(q-1)(q-2)q^7, & k \text{ even}. \end{cases}$$

6.4. Exact evaluation and the conjecture. Every W, W' is the exponential sum of an $\mathbb{F}_2$ -quadratic form (plus a linear form) on $\mathbb{F}_2^{12k}$, hence computable exactly by linear algebra: radical, test of $Q + L$ on it, a symplectic basis of a complement and its Arf invariant. We implemented this (`legendre_d6_qf.py`) with $\mathbb{F}_{q^{12}} = \mathbb{F}_q[z]/(m)$ and the maps $\beta \mapsto \beta^2, \beta^{1/2}, \sigma, \theta$ as $12k \times 12k$ matrices over $\mathbb{F}_2$.

Conjecture 6.5. For all $q = 2^k$: $A_6 = q^9 + \varepsilon_0(q-1)q^5$ and $X = \varepsilon_0 q^6 + q^5$, i.e.

$$\sum_{\lambda_5 \neq 0} W(0, \lambda_5) + \sum_{\lambda_3 \lambda_5 \neq 0} W(\lambda_3, \lambda_5) = (q-1)q^7(\varepsilon_0 q + 2), \quad \sum_{\lambda_3, \lambda_5} W'(\lambda_3, \lambda_5) = q^3(\varepsilon_0 q^6 + q^5).$$

The statement for A_6 with k odd is proved below (Corollary 6.13); the statement for X , and that for A_6 with k even, remain conjectural.

Proposition 6.6 (structure of the data). For $k \leq 7$ the following hold.

- (a) k odd: $W(0, \lambda_5) = 2q^7$ for all $\lambda_5 \neq 0$; the mixed W depends only on $j = \lambda_3^5 / \lambda_5^3 \in \mathbb{F}_q^\times$ and equals $2q^7$ for exactly $(q-2)/2$ values of j and 0 otherwise. Hence $\sum W = (q-1)2q^7 + \frac{(q-1)(q-2)}{2} 2q^7 = (q-1)q^8$.

TABLE 1. Exact values of A_6 and X by the quadratic-form algorithm ($N = 12k$ bits). For $k \leq 5$ the values coincide with the Hayes-group FFT of [4]; $k = 6, 7$ are beyond its reach at $d = 6$. In all cases $A_6 = q^9 + (-1)^{k+1}(q-1)q^5$ and $X = (-1)^{k+1}q^6 + q^5$.

k	q	N	A_6	X	time
1	2	12	544	96	< 1 s
2	4	24	259 072	-3 072	< 1 s
3	8	36	134 447 104	294 912	< 1 s
4	16	48	68 703 748 096	-15 728 640	2 s
5	32	60	35 185 412 276 224	1 107 296 256	9 s
6	64	72	18 014 330 863 747 072	-67 645 734 912	48 s
7	128	84	9 223 376 400 541 548 544	4 432 406 249 472	225 s

TABLE 2. Distribution of $W(\lambda_3, \lambda_5)/q^7$ over the $q^2 - 1$ nontrivial pairs (radical dimension in parentheses); “pure cubic” $\lambda_5 = 0$, “pure quintic” $\lambda_3 = 0$, “mixed” $\lambda_3\lambda_5 \neq 0$.

k	pure cubic ($q-1$)	pure quintic ($q-1$)	mixed $((q-1)^2)$
1	0 (4)	2 (4)	0 (6) $\times 1$
3	0 (8)	2 (8)	2 (8) $\times 21$; 0 (10) $\times 28$
5	0 (12)	2 (12)	2 (12) $\times 465$; 0 (14) $\times 496$
7	0 (16)	2 (16)	2 (16) $\times 8001$; 0 (18) $\times 8128$
2	-2 (6)	0 (8)	-4 (8) $\times 3$; 1 (4) $\times 6$
4	-2 (10)	-4 (12) $\times 3$; 1 (8) $\times 12$	-4 (12) $\times 75$; 1 (8) $\times 90$; 0 (12) $\times 60$
6	-2 (14)	0 (16)	-4 (16) $\times 1386$; 1 (12) $\times 1638$; 0 (16) $\times 945$

- (b) k even: $W(0, \lambda_5) = 0$ if $k \equiv 2 \pmod{4}$, while for $k \equiv 0 \pmod{4}$ (so $5 \mid q-1$) it equals $-4q^7$ for the $(q-1)/5$ fifth powers and $+q^7$ otherwise; the mixed W take values $-4q^7, q^7, 0$ with multiplicities $(q-1)(n_-, n_+, n_0)$, $(n_-, n_+, n_0) = (1, 2, 0), (5, 6, 4), (22, 26, 15)$ for $k = 2, 4, 6$, satisfying $n_+ - 4n_- = 2 - q$.

The set of j in (a) is not of the form $\{\text{Tr}_{q/2}(j^m) = c\}$ for any m when $k = 5, 7$ (it is $\{\text{Tr } j = 0\}$ when $k = 3$).

6.5. The mixed part for odd k : a Gold-type quintic and a $\mathbb{Z}/4$ condition. For k odd we describe the mixed terms completely in terms of the factorisation of a single quintic over $\mathbb{F}_q$; the evaluation is completed in Theorem 6.12 below.

Proposition 6.7. *Let k be odd, $\lambda_3\lambda_5 \neq 0$, $j = \lambda_3^5/\lambda_5^3$, $t = j^{1/5} \in \mathbb{F}_q$ and $c = j^{-1/2} \in \mathbb{F}_q^\times$ (fifth and square roots are unique in $\mathbb{F}_q$).*

- (a) $W(\lambda_3, \lambda_5)$ depends only on j (the orbits of $(\lambda_3, \lambda_5) \mapsto (s^3\lambda_3, s^5\lambda_5)$, $s \in \mathbb{F}_q^\times$, are the fibres of j).
- (b) With $y = \lambda_5^{-1/5}u$, the kernel equation $\lambda_5^4y^{15} + \lambda_3^4y^7 + \lambda_3^2y + \lambda_5 = 0$ becomes $P_j(u) := u^{15} + t^4u^7 + t^2u + 1 = 0$, and

$$P_j(u) = (u^5 + t^2u + 1)(u^{10} + t^2u^6 + u^5 + 1), \quad u = t^{1/2}v \implies u^5 + t^2u + 1 = t^{5/2}(v^5 + v + c).$$

The five roots $u_0, \dots, u_4$ of $u^5 + t^2u + 1$ have sum 0, are $\mathbb{F}_2$ -linearly independent apart from this relation, and $\ker L_\lambda = \{0\} \cup \{u_i\} \cup \{u_i + u_{i'}\}$ (the ten roots of the second factor are the pairwise sums). The q -Frobenius acts on $\ker L_\lambda$ through its permutation action on the roots v_i of $v^5 + v + c$, i.e. through the standard embedding $S_5 \rightarrow GL_4(\mathbb{F}_2)$ on the sum-zero module.

- (c) Let $g \in S_5$ be the Frobenius permutation of the roots of $v^5 + v + c$ (its cycle type is the factorisation type of $v^5 + v + c$ over $\mathbb{F}_q$). Then $\ker L_\lambda \subset \mathbb{F}_{q^{12}}$ iff g has no 5-cycle, $\dim_{\mathbb{F}_2}(\ker L_\lambda \cap \text{im } \theta) = 2$ if g is a 4-cycle (type $(1, 4)$) and $= 4$ for types $(1, 1, 1, 2)$, $(2, 3)$, $(1, 1, 1, 1, 1)$, $(1, 2, 2)$, $(1, 1, 3)$, and the image of $\text{Tr}_{q^{12}/q}$ on $\ker L_\lambda$ is one-dimensional over $\mathbb{F}_2$ if g is a 4-cycle and zero otherwise. Consequently, for a 4-cycle, $|W| \in \{0, 2q^7\}$ and exactly $q/2$ of the sums $S(\lambda_1, \lambda_3, \lambda_5)$ are nonzero (with modulus $4q^6$); for the other types $|W| \in \{0, 4q^7\}$ and either all or none of the $S(\lambda_1, \cdot)$ vanish.

Proof. (a) For $s \in \mathbb{F}_q^\times$, $\beta \mapsto s\beta$ sends $\text{Tr}(\lambda_3\beta^2\theta\beta + \lambda_5\beta^4\theta\beta)$ to the form with $(s^3\lambda_3, s^5\lambda_5)$; since $3, 5 \nmid q - 1$ the map $s \mapsto (s^3, s^5)$ is injective, so the orbits have size $q - 1$ and are the fibres of j . (b) Substitute $y = \lambda_5^{-1/5}u$ and divide by λ_5 . The factorisation is checked by expanding: $u^5 P_j(u) = (u^{10} + t^2 u^6 + u^5)^2 + (u^{10} + t^2 u^6 + u^5)$. The second factor equals $u^5 A(u) + 1$ with $A = u^5 + t^2 u + 1$, so its roots are not roots of A ; A is separable ($A' = (u + \sqrt{t})^4$, $A(\sqrt{t}) = 1$) with $e_1 = 0$; $\ker L_\lambda$ is a 4-dimensional $\mathbb{F}_2$ -space containing the u_i , and a relation $u_i + u_{i'} + u_{i''} = 0$ would force two of the remaining roots to coincide, while $u_i + u_{i'} + u_{i''} + u_{i'''} = 0$ would force the fifth root to be 0; hence the pairwise sums are 10 distinct nonzero kernel elements outside $\{u_i\}$, which exhausts the kernel. Frobenius commutes with the identification. (c) Rationality over $\mathbb{F}_{q^{12}}$ means $g^{12} = 1$. On the sum-zero module $V \cong \mathbb{F}_2[\mathbb{Z}/4]$ of a 4-cycle, $\sum_{i < 12} g^i = 3(1 + g + g^2 + g^3) = (1 + g)^3$ has rank 1 and $\sum_{i < 6} g^{2i} = 1 + g^2 = (1 + g)^2$ has kernel of dimension 2; for g of order 1, 2, 3, 6 both operators vanish on V (the sums have an even number of equal terms), giving trace image 0 and $\ker \cap \text{im } \theta = \ker$. The last sentence follows from the radical dimensions $2k + \dim(\ker \cap \text{im } \theta)$ for W and $\dim(\ker L_\lambda) = 4$ for the $S(\lambda_1, \cdot)$, together with the criterion $S \neq 0$ iff $\text{Tr}(\lambda_1 \cdot) = Q_x$ on $\ker L_\lambda$: for a 4-cycle the functionals $\text{Tr}(\lambda_1 \cdot)|_{\ker}$ form a one-dimensional space and $Q_x|_{\ker}$ lies in it (both are determined by whether $\text{Tr}_{q^{12}/q}(x)$ vanishes, as Q_x is additive on the radical and takes the same value on the four roots of degree 4), so exactly half of the λ_1 qualify. $\square$

Proposition 6.8 (the quintic $v^5 + v + c$, k odd). *Let k be odd and $c \in \mathbb{F}_q^\times$. The factorisation type of $v^5 + v + c$ over $\mathbb{F}_q$ is one of $(1, 4)$, $(2, 3)$, $(1, 1, 1, 2)$, and these occur for exactly*

$$N_{(1,4)} = \frac{q-2}{2}, \quad N_{(2,3)} = \frac{q+1}{3}, \quad N_{(1,1,1,2)} = \frac{q-2}{6}$$

values of c . In particular the number of $\mathbb{F}_q$ -roots is 1, 0 or 3 with these multiplicities, in agreement with the theorem of Helleseth–Kholosha [9] on $x^{2^l+1} + x + a$, $\gcd(l, k) = 1$ (here $l = 2$), and the quintic is never irreducible.

Proof. Put $t = c^{-2/5}$ (a bijection of $\mathbb{F}_q^\times$) and $A(u) = u^5 + t^2 u + 1$, so that $u = t^{1/2}v$ identifies the roots of $v^5 + v + c$ with those of A (Proposition 6.7(b)) and $K = \ker L_\lambda \subset \mathbb{F}_q$, $\lambda = (\cdot, t, 1)$, is the sum-zero $\mathbb{F}_2$ -span of the five roots $u_0, \dots, u_4$ of A , on which the q -Frobenius σ acts through its permutation $g \in S_5$ of the roots.

Only three types occur. The function $x \mapsto \text{Tr}_{q/2}(x^5 + tx^3)$ on $\mathbb{F}_q$ is an $\mathbb{F}_2$ -quadratic form with polar form $\text{Tr}_{q/2}(y T_\lambda(x))$, whose radical is $\{x \in \mathbb{F}_q : T_\lambda(x) = 0\} = K^\sigma$, the σ -fixed subspace of K . Since the polar form is alternating and nondegenerate on $\mathbb{F}_q/K^\sigma$, $\dim K^\sigma \equiv k \equiv 1 \pmod{2}$. On the sum-zero module of S_5 the fixed space of g has dimension 1, 1, 3 for the cycle types $(1, 4)$, $(2, 3)$, $(1, 1, 1, 2)$ and 0, 2, 2, 4 for (5) , $(1, 1, 3)$, $(1, 2, 2)$, $(1, 1, 1, 1, 1)$ (count the g -stable subsets of $\{0, \dots, 4\}$ modulo complementation). Hence only the first three types occur.

Counting. (i) The map $v \mapsto v^5 + v$ sends $\mathbb{F}_q$ onto the values c , with $v^5 + v = 0$ iff $v \in \{0, 1\}$ (as $v^4 = 1$ forces $v = 1$); so $N_{(1,4)} + 3N_{(1,1,1,2)} = q - 2$. (ii) Let $v \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$, $\bar{v} = v^q$, $s = v + \bar{v} \neq 0$, $n = v\bar{v}$. Then $\bar{v}^5 + \bar{v} + v^5 + v = (v + \bar{v})(s^4 + ns^2 + n^2 + 1)$, so $v^5 + v \in \mathbb{F}_q$ iff $s^4 + ns^2 + n^2 = 1$. Writing $n = s^2 m$, irreducibility of $v^2 + sv + n$ means $\text{Tr}_{q/2}(m) = 1$, and the condition becomes $m^2 + m = 1 + s^{-4}$, which is solvable iff $\text{Tr}_{q/2}(1 + s^{-4}) = 0$ iff $\text{Tr}_{q/2}(s^{-1}) = 1$ (k odd), in which case exactly one of the two solutions $m, m + 1$ has $\text{Tr } m = 1$. Hence exactly $2 \cdot \#\{s \in \mathbb{F}_q^\times : \text{Tr}(1/s) = 1\} = q$ elements $v \in \mathbb{F}_{q^2} \setminus \mathbb{F}_q$ satisfy $v^5 + v \in \mathbb{F}_q$, necessarily in $\mathbb{F}_q^\times$. A quintic of type $(1, 4)$, $(2, 3)$, $(1, 1, 1, 2)$ has 0, 2, 2 roots in $\mathbb{F}_{q^2} \setminus \mathbb{F}_q$, so $2N_{(2,3)} + 2N_{(1,1,1,2)} = q$. (iii) $N_{(1,4)} + N_{(2,3)} + N_{(1,1,1,2)} = q - 1$. Solving (i)–(iii) gives the three counts. $\square$

6.6. The mixed part for odd k : the theorem. Throughout this subsection k is odd, $(\lambda_3, \lambda_5) = (t, 1)$ with $t \in \mathbb{F}_q^\times$ (every class $j = t^5$ has such a representative, Proposition 6.7(a)), $A(u) = u^5 + t^2u + 1$ with roots $u_0, \dots, u_4$, $K = \ker L_\lambda = \ker T_\lambda \subset \mathbb{F}_{q^{12}}$ (Proposition 6.7(b)), $T = T_\lambda$, σ is the q -Frobenius and $\tau = \text{Tr}_{q^{12}/q}$. For $b \in \mathbb{F}_q$ put

$$S(b) = S(b, t, 1) = \sum_{x \in \mathbb{F}_{q^{12}}} \psi_0(x^5 + tx^3 + bx), \quad Q_b(x) = \text{Tr}_{q^{12}/2}(x^5 + tx^3 + bx),$$

so that $W(t, 1) = \sum_{b \in \mathbb{F}_q} S(b)$ by (7), and $S(b) \in \{0, \pm 4q^6\}$ since $\dim K = 4$.

Lemma 6.9 (shifts). *Let $V = T^{-1}(\mathbb{F}_q) = \{c \in \mathbb{F}_{q^{12}} : T(c) \in \mathbb{F}_q\}$.*

- (a) *For $c \in V$ and $b \in \mathbb{F}_q$: $S(b) = (-1)^{Q_b(c)} S(b + T(c))$.*
- (b) *$c \in V$ iff $(\sigma - 1)c \in K$, and then $\tau(c) = (1 + \sigma^2 + \sigma^4 + \dots + \sigma^{10})(\sigma - 1)c$.*
- (c) *If $c \in V$ and $\tau(c) = 0$, then $W(t, 1) = (-1)^{Q_0(c)} W(t, 1)$; in particular $W(t, 1) = 0$ if $Q_0(c) = 1$.*

Proof. (a) $Q_b(x + c) = Q_b(x) + Q_b(c) + \text{Tr}(xT(c))$ and $\text{Tr}_{q^{12}/2}(xT(c)) = \text{Tr}_{q^{12}/2}(T(c)x)$ with $T(c) \in \mathbb{F}_q$; substitute $x \mapsto x + c$ in $S(b)$. (b) L_λ has coefficients in $\mathbb{F}_q$ and commutes with σ , so $L_\lambda(c) \in \mathbb{F}_q$ iff $L_\lambda((\sigma - 1)c) = 0$; and $T(c) \in \mathbb{F}_q$ iff $T(c)^4 = L_\lambda(c) \in \mathbb{F}_q$. With $c_0 = (\sigma - 1)c$ one has $\sigma^i c = c + \sum_{m < i} \sigma^m c_0$, hence $\tau(c) = 12c + \sum_{m \leq 10} (11 - m)\sigma^m c_0 = \sum_{m \text{ even}} \sigma^m c_0$. (c) Sum (a) over $b \in \mathbb{F}_q$: $Q_b(c) = Q_0(c) + \text{Tr}_{q/2}(b\tau(c)) = Q_0(c)$, and $b \mapsto b + T(c)$ permutes $\mathbb{F}_q$. $\square$

Lemma 6.10 (a quadratic factor forces $W = 0$). *Suppose $A(u)$ has an irreducible quadratic factor $u^2 + \kappa u + N$ over $\mathbb{F}_q$ (types (2, 3) and (1, 1, 1, 2)). Then $N = \kappa^{-3}$, $t = \kappa^2 + \kappa^{-1/2} + \kappa^{-3}$ and $\text{Tr}_{q/2}(\kappa^{-5}) = 1$. Let u be a root of the quadratic factor and $c \in \mathbb{F}_{q^4}$ with $c^q + c = u$. Then $c \in V$, $\tau(c) = 0$ and $Q_0(c) = \text{Tr}_{q/2}(\kappa^{-5}) = 1$. Consequently $W(t, 1) = 0$. Moreover every $S(b)$, $b \in \mathbb{F}_q$, equals $\pm 4q^6$, and $S(b) = -S(b + T(c))$: the q sums cancel in pairs.*

Proof. Dividing, $A = (u^2 + \kappa u + N)(u^3 + \kappa u^2 + (\kappa^2 + N)u + \kappa^3)$ with $N\kappa^3 = 1$ and $t^2 = \kappa^4 + N\kappa^2 + N^2 = \kappa^4 + \kappa^{-1} + \kappa^{-6}$; the quadratic is irreducible iff $\text{Tr}_{q/2}(N/\kappa^2) = \text{Tr}_{q/2}(\kappa^{-5}) = 1$. Since $u \in \mathbb{F}_{q^2}$, $\text{Tr}_{q^4/q}(u) = 2(u + u^q) = 0$ and c exists. By Lemma 6.9(b), $(\sigma - 1)c = u \in K$ gives $c \in V$, and $\tau(c) = (1 + \sigma^2 + \dots + \sigma^{10})u = 6u = 0$. The conjugates of c over $\mathbb{F}_q$ are $c, c + u, c + \kappa, c + u'$ with $u' = u^q = u + \kappa$. As $[\mathbb{F}_{q^{12}} : \mathbb{F}_{q^4}] = 3$ is odd, $Q_0(c) = \text{Tr}_{q^4/2}(c^5 + tc^3) = \text{Tr}_{q/2} \sum_i (c_i^5 + tc_i^3)$ over the four conjugates $c_i = c + a_i$, $a_i \in \{0, u, \kappa, u'\}$. In characteristic 2, $(c + a)^3 = c^3 + c^2a + ca^2 + a^3$ and $(c + a)^5 = c^5 + c^4a + ca^4 + a^5$, and $\sum a_i = \sum a_i^2 = \sum a_i^4 = 0$, so $\sum_i c_i^3 = \sum a_i^3 = u^3 + u'^3 + \kappa^3 = \kappa N = \kappa^{-2}$ and $\sum_i c_i^5 = u^5 + u'^5 + \kappa^5 = \kappa^3 N + \kappa N^2 = 1 + \kappa^{-5}$ (Newton's identities for $u + u' = \kappa$, $uu' = N$). Hence

$$Q_0(c) = \text{Tr}_{q/2}(1 + \kappa^{-5} + t\kappa^{-2}), \quad t\kappa^{-2} = 1 + \kappa^{-5/2} + \kappa^{-5},$$

and $\text{Tr}(1) = 1$, $\text{Tr}(\kappa^{-5/2}) = \text{Tr}(\kappa^{-5})$ give $Q_0(c) = \text{Tr}_{q/2}(\kappa^{-5}) = 1$. Lemma 6.9(c) gives $W = 0$. For the last assertion: by Proposition 6.7(c) the trace τ vanishes on K in these types, so $\text{Tr}_{q^{12}/2}(bx) = 0$ for $x \in K$; and Q_0 vanishes on K because it is additive there and each root u_i lies in a subfield $\mathbb{F}_q, \mathbb{F}_{q^2}$ or $\mathbb{F}_{q^3}$ of even index in $\mathbb{F}_{q^{12}}$. By Theorem 6.1, $S(b) \neq 0$ for all b , and Lemma 6.9(a) with $Q_b(c) = 1$ gives the pairing (note $T(c) \notin T(\mathbb{F}_q)$, since otherwise $c \in \mathbb{F}_q + K$ and $Q_0(c)$ would vanish). $\square$

Lemma 6.11 (the 4-cycle: vanishing over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ fixes the sign). *Suppose A has type (1, 4): $u_0 \in \mathbb{F}_q$ and σ cycles $u_1 \rightarrow u_2 \rightarrow u_3 \rightarrow u_4 \rightarrow u_1$. Put $\kappa = 1 + \text{Tr}_{q/2}(u_0^5 + tu_0^3) \in \mathbb{F}_2$ and, for $b \in \mathbb{F}_q$ and $m \geq 1$, $s_m(b) = \sum_{x \in \mathbb{F}_{q^m}} \psi_0(x^5 + tx^3 + bx)$, so that $S(b) = s_{12}(b)$.*

- (a) *$S(b) \neq 0$ iff $\text{Tr}_{q/2}(bu_0) = \kappa$ iff $s_1(b) = 0$; this holds for exactly $q/2$ values of b .*
- (b) *If $s_1(b) = 0$ then $s_2(b) = 0$.*
- (c) *For these b the L -polynomial of $C_b : y^2 + y = x^5 + tx^3 + bx$ over $\mathbb{F}_q$ is $1 + q^2T^4$, hence $S(b) = +4q^6$. Therefore $W(t, 1) = \frac{q}{2} \cdot 4q^6 = 2q^7$.*

Proof. (a) By Theorem 6.1, $S(b) \neq 0$ iff $\text{Tr}_{q^{12/2}}(bx) = Q_0(x)$ for all $x \in K$; both sides are additive on K . For $x = u_0 \in \mathbb{F}_q$ both sides vanish ($\text{Tr}_{q^{12/2}}$ is $12 \cdot \text{id}$ on $\mathbb{F}_q$). For $i \geq 1$, $u_i \in \mathbb{F}_{q^4}$ and $\tau(u_i) = 3 \text{Tr}_{q^4/q}(u_i) = u_1 + u_2 + u_3 + u_4 = u_0$, so $\text{Tr}_{q^{12/2}}(bu_i) = \text{Tr}_{q/2}(bu_0)$, while $Q_0(u_i) = \text{Tr}_{q/2}(\sum_{i' \geq 1} (u_{i'}^5 + tu_{i'}^3)) = \text{Tr}_{q/2}(p_5 - u_0^5 + t(p_3 - u_0^3))$ with p_m the power sums of all five roots of A ; Newton's identities for A ($e_1 = e_2 = e_3 = 0$, $e_4 = t^2$, $e_5 = 1$) give $p_3 = 0$, $p_5 = 5e_5 = 1$, so $Q_0(u_i) = \text{Tr}_{q/2}(1 + u_0^5 + tu_0^3) = \kappa$. Thus $S(b) \neq 0$ iff $\text{Tr}_{q/2}(bu_0) = \kappa$ iff $\text{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) = 1$. On the other hand $x \mapsto \text{Tr}_{q/2}(x^5 + tx^3 + bx)$ is a quadratic form on $\mathbb{F}_q$ with polar form $\text{Tr}_{q/2}(yT(x))$ and radical $K^\sigma = \{0, u_0\}$, so $s_1(b) \neq 0$ iff $\text{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) = 0$.

(b) Over $\mathbb{F}_{q^2}$ the radical is $K^{\sigma^2} = \{0, u_0, w, w'\}$ with $w = u_1 + u_3$, $w' = \sigma w = u_2 + u_4 = w + u_0$, and $s_2(b) \neq 0$ iff $\text{Tr}_{q^2/2}(x^5 + tx^3 + bx)$ vanishes at $x = u_0$ and $x = w$. At u_0 it does. Let $N_2 = ww' \in \mathbb{F}_q$ and let $\rho \in \mathbb{F}_q$ be the cube root of u_0 ($3 \nmid q - 1$). We claim $N_2 = \rho$. Dividing A by $u + u_0$ and using $u_0^5 + t^2u_0 + 1 = 0$, the quartic with roots $u_1, \dots, u_4$ is $B(u) = u^4 + u_0u^3 + u_0^2u^2 + u_0^3u + u_0^{-1}$; its resolvent cubic, with roots $u_1u_2 + u_3u_4$, $u_1u_3 + u_2u_4$, $u_1u_4 + u_2u_3$, is $\theta^3 + u_0^2\theta^2 + u_0^4\theta + u_0^6 + u_0 = (\theta + u_0^2)^3 + u_0$, whose unique root in $\mathbb{F}_q$ is $u_0^2 + \rho$; the σ -invariant root is $M = u_1u_3 + u_2u_4$, so $M = u_0^2 + \rho$ and $N_2 = (u_1 + u_3)(u_2 + u_4) = e_2(B) + M = u_0^2 + M = \rho$. Now with $w + w' = u_0$, $ww' = \rho$: $w^3 + w'^3 = u_0^3 + u_0\rho$ and $w^5 + w'^5 = u_0^5 + u_0^3\rho + u_0\rho^2$, so

$$\begin{aligned} \text{Tr}_{q^2/2}(w^5 + tw^3 + bw) &= \text{Tr}_{q/2}(u_0^5 + tu_0^3 + bu_0) + \text{Tr}_{q/2}(u_0^3\rho + u_0\rho^2 + tu_0\rho) \\ &= 1 + \text{Tr}_{q/2}(\rho^{10} + \rho^5 + t\rho^4) = 1 + \text{Tr}_{q/2}(t\rho^4). \end{aligned}$$

From $u_0^5 + t^2u_0 + 1 = 0$, $t^2 = u_0^4 + u_0^{-1} = \rho^{12} + \rho^{-3}$, so $t = \rho^6 + \rho^{-3/2}$ and $t\rho^4 = \rho^{10} + \rho^{5/2}$, whose trace is $\text{Tr}(\rho^5) + \text{Tr}(\rho^5) = 0$. Hence the form takes the value 1 at w and $s_2(b) = 0$.

(c) With $\omega_1, \dots, \omega_4$ the Frobenius eigenvalues of C_b (genus 2), $s_m(b) = -\sum_i \omega_i^m$ (Theorem 6.1), so $e_1 = \sum \omega_i = -s_1 = 0$ and $e_2 = \frac{1}{2}(e_1^2 - \sum \omega_i^2) = \frac{1}{2}(s_1^2 + s_2) = 0$; the functional equation gives $e_3 = qe_1 = 0$, $e_4 = q^2$. So $\prod(1 - \omega_i T) = 1 + q^2 T^4$, $\omega_i^4 = -q^2$, $\omega_i^{12} = -q^6$ and $S(b) = -\sum \omega_i^{12} = 4q^6$. Summing over the $q/2$ values of b in (a) gives $W(t, 1) = 2q^7$. $\square$

Theorem 6.12 (the mixed part, k odd). *Let k be odd, $\lambda_3\lambda_5 \neq 0$, $j = \lambda_3^5/\lambda_5^3$ and $c = j^{-1/2}$. Then*

$$W(\lambda_3, \lambda_5) = \begin{cases} 2q^7, & v^5 + v + c \text{ has type } (1, 4), \\ 0, & v^5 + v + c \text{ has type } (2, 3) \text{ or } (1, 1, 1, 2), \end{cases}$$

and no other type occurs. In the 4-cycle case exactly $q/2$ of the sums $S(\lambda_1, \lambda_3, \lambda_5)$, $\lambda_1 \in \mathbb{F}_q$, are nonzero and each equals $\pm 4q^6$; in the other two cases all q sums equal $\pm 4q^6$ and cancel in pairs. Equivalently, in the β -form of (7): Q vanishes on the radical $\theta^{-1}(\ker L_\lambda \cap \mathbb{F}_{q^2})$ (of dimension $2k + 2$) with Arf sign $+$ in the 4-cycle case, and Q does not vanish on the radical (of dimension $2k + 4$) otherwise.

Proof. By Proposition 6.7(a),(b) we may take $(\lambda_3, \lambda_5) = (t, 1)$, $t = j^{1/5}$, and the type of $v^5 + v + c$ is the type of $A(u) = u^5 + t^2u + 1$. Proposition 6.8 restricts the types to the three listed (the scaling $(\lambda_1, \lambda_3, \lambda_5) \mapsto (s\lambda_1, s^3\lambda_3, s^5\lambda_5)$ preserves the number and signs of the nonzero S). Lemma 6.10 covers types (2, 3) and (1, 1, 1, 2), Lemma 6.11 the type (1, 4). For the β -form: by Proposition 6.7(c) the radical has dimension $2k + 2$ or $2k + 4$, so $|W| \in \{0, 2q^7\}$ or $\{0, 4q^7\}$ respectively, and $W = 2q^7$ (resp. $W = 0$) is equivalent to the stated behaviour of Q on the radical. $\square$

Corollary 6.13. *For all odd k : $\sum_{\lambda_3\lambda_5 \neq 0} W(\lambda_3, \lambda_5) = (q - 1)(q - 2)q^7$ and*

$$A_6 = |U| = q^9 + (q - 1)q^5.$$

That is, Conjecture 6.5 holds for A_6 whenever k is odd.

Proof. Each class $j \in \mathbb{F}_q^\times$ consists of $q - 1$ pairs (λ_3, λ_5) (Proposition 6.7(a)), and the 4-cycle classes number $(q - 2)/2$ (Proposition 6.8); Theorem 6.12 gives $\sum_{\text{mixed}} W = (q - 1)\frac{q-2}{2} 2q^7$, which is (9) for odd k . Combine with Lemmas 6.2, 6.4 and (7). $\square$

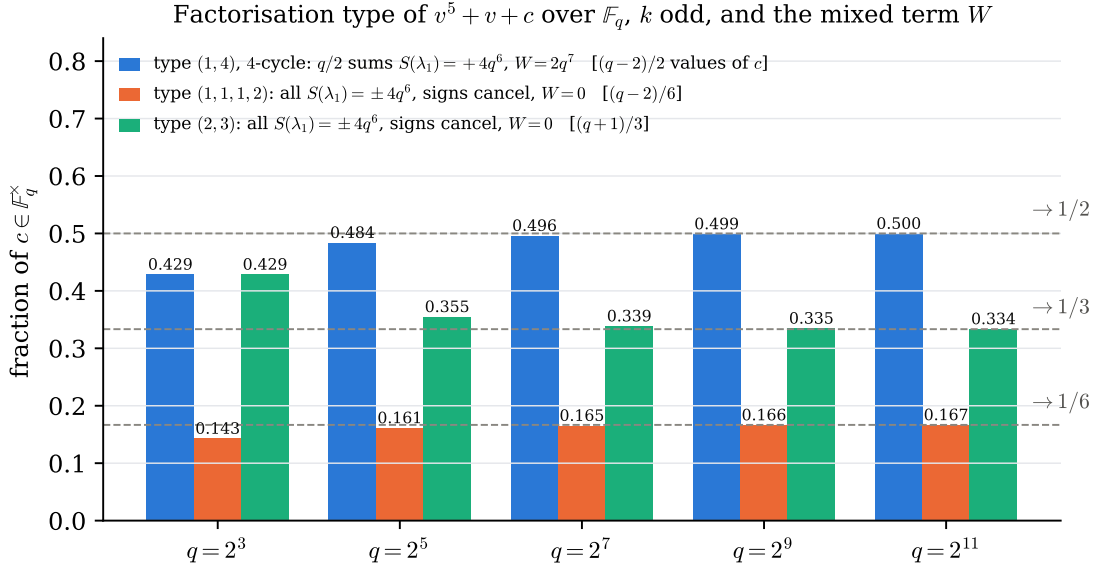


FIGURE 2. The mixed part for odd k (Theorem 6.12). For each $c \in \mathbb{F}_q^\times$ the Gold-type quintic $v^5 + v + c$ has one of three factorisation types over $\mathbb{F}_q$, with the exact frequencies of Proposition 6.8 (bars: $q = 2^3, \dots, 2^{11}$; dashed lines: the limits $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$). Only the 4-cycle type contributes to A_6 : there $q/2$ of the q curve sums $S(\lambda_1)$ equal $+4q^6$ and the rest vanish; for the other types all q sums are $\pm 4q^6$ and cancel in pairs (Lemmas 6.10, 6.11). The “half-measure” of the 4-cycle set, $(q-2)/2$, is what produces the term $(q-1)(q-2)q^7$ of (9).

Theorem 6.12 was checked in two independent ways: for $k \leq 7$ by the exact $\mathbb{F}_2$ -quadratic-form algorithm (`legendre_d6_qf_mixed.py`: every j , radical test and signs), and for $k \leq 9$ through the L -polynomials (`kasami_sign.py`): for every $j \in \mathbb{F}_q^\times$ and every $b \in \mathbb{F}_q$ the sums $s_1(b), s_2(b)$ over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$ determine $\prod(1 - \omega_i T)$ and hence $S(b) = -\sum \omega_i^{12}$; the resulting multisets $\{S(b)\}$ and the values of W agree with the theorem in every case, and in the 4-cycle case the predicate $\text{Tr}_{q/2}(bu_0) = \kappa$ of Lemma 6.11(a) matches $S(b) \neq 0$ exactly. The types of Proposition 6.8 were enumerated for $k \leq 11$ (`quintic_check.py`).

The condition “ g is a 4-cycle” is a $\mathbb{Z}/4$ -condition (the Frobenius must be a regular unipotent on the kernel), not an $\mathbb{F}_2$ -trace condition; this is why no predicate $\text{Tr}(f(j)) = 0$ describes the set of j with $W \neq 0$ (Proposition 6.6). Note also that the sign in the 4-cycle case is *not* obtained from an Arf-invariant computation on $\mathbb{F}_2^{12k}$ but from the two small sums s_1, s_2 over $\mathbb{F}_q$ and $\mathbb{F}_{q^2}$, which vanish for exactly the b with $S(b) \neq 0$ and force the Frobenius eigenvalues of C_b to be $\sqrt{q}$ times primitive 8th roots of unity. For even k the orbit invariant j is not complete (cube and fifth roots need not exist in $\mathbb{F}_q$) and the analogous analysis has not been carried out; the closed form of X (the sums W') is open for all k .

7. THE SCALING LAW AT $d = 6$ AND THE NATURE OF Y

7.1. Variance formula.

Theorem 7.1. *Assume Conjecture 6.5 (true for $q \leq 128$; for odd k only its statement for X is needed, by Corollary 6.13) and write $Y = q^3 m_k$. Then $B = -2q^3 [k \text{ odd}]$,*

$$\varphi_0 = \begin{cases} q^5 + q^4 - 2q^3, & k \text{ odd}, \\ -q^5 + q^4, & k \text{ even}, \end{cases} \quad \varphi_{A,B} = \begin{cases} -2q^3 \pm q^2 m_k, & k \text{ odd}, \\ \pm q^2 m_k, & k \text{ even}, \end{cases}$$

and

$$\frac{\text{Var}_{\mathcal{G}^2}(\Psi)}{q^7} = \begin{cases} \frac{(q-1)^2(q+2)^2}{q^2} + \frac{(q-1)(4q^2 + m_k^2)}{q^4}, & k \text{ odd}, \\ (q-1)^2 + \frac{(q-1)m_k^2}{q^4}, & k \text{ even}. \end{cases}$$

In particular $\text{Var}_{\mathcal{G}^2}(\Psi)/q^7 = q^2(1 + O(1/q))$, i.e. $C_6 := \lim_{q \rightarrow \infty} \text{Var}_{\mathcal{G}^2}(\Psi)/q^9 = 1$, provided $m_k = o(q^{5/2})$.

Proof. Substitute into Corollary 4.3. □

TABLE 3. $d = 6$: exact variances (Hayes FFT, $q \leq 32$; $q = 16$ also by CRT), the three Legendre deviations and $m_k = Y/q^3$. The last two columns show that $\text{Var}_{\mathcal{G}^2}/q^7 \asymp q^2$ and not q^4 .

q	$\text{Var}_{\mathcal{G}}/q^7$	$\text{Var}_{\mathcal{G}^2}/q^7$	ratio	φ_0	φ_A	φ_B	m_k	$\text{Var}_{\mathcal{G}^2}/q^9$
2	5.7578	15.5625	2.703	32	36	-68	13	3.891
4	3.7004	9.0117	2.435	-768	-16	16	-1	0.563
8	4.7483	79.0935	16.657	35 840	-3 264	1 216	-35	1.236
16	4.3514	225.5056	51.824	-983 040	12 032	-12 032	47	0.881
32	4.2086	1100.8844	261.581	34 537 472	685 056	-816 128	733	1.075

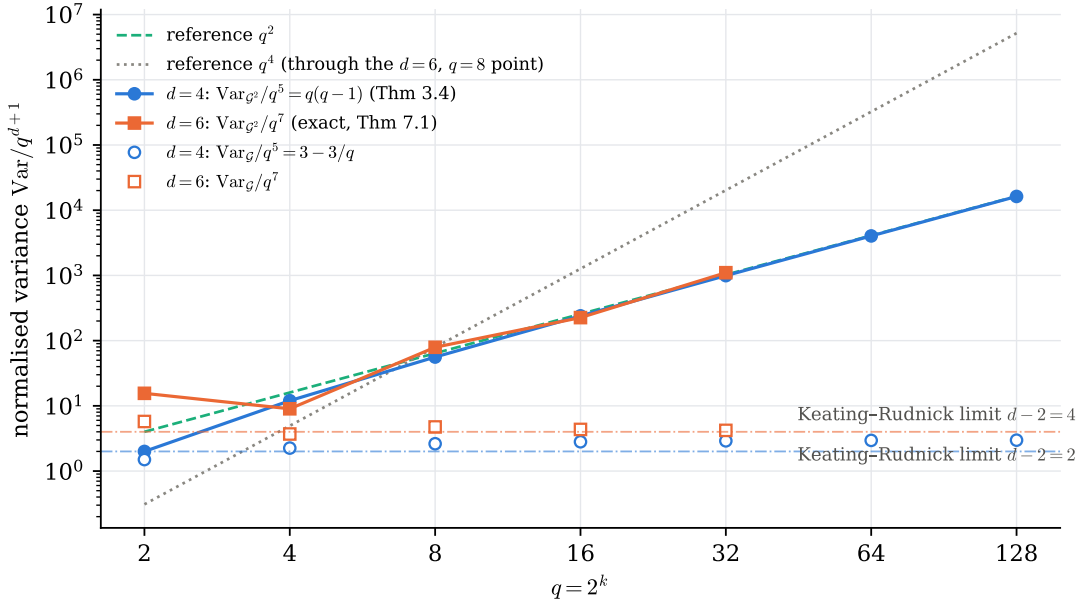


FIGURE 3. Normalised variances against q (log-log). Filled markers: $\text{Var}_{\mathcal{G}^2}(\Psi)/q^{d+1}$, exact values ($d = 4$: $q(q-1)$ for $q \leq 128$, Theorem 3.4; $d = 6$: Table 3). Open markers: the full-group variances $\text{Var}_{\mathcal{G}}/q^{d+1}$, which approach 3 ($d = 4$, Theorem 3.11) and the Keating–Rudnick value 4 ($d = 6$). The dashed line is q^2 ; the dotted line is a q^4 law normalised through the $d = 6$, $q = 8$ point, which the data exclude. Whether $\lim \text{Var}_{\mathcal{G}^2}/q^9 = 1$ at $d = 6$ depends on $m_k = o(q^{5/2})$ (Theorem 7.1).

Remark 7.2. At $d = 4$ the Legendre anomaly is a single class with relative deficit $1/q$; at $d = 6$ it is the q classes with $c_2 = 0$ (the intervals around f^2 with $f_5 = 0$), each with relative deviation $\pm q^{-2}$ and sign ε_0 . Both mechanisms give $\text{Var}_{\mathcal{G}^2}/q^{d+1} \asymp q^2$. The full-group variance $\text{Var}_{\mathcal{G}}/q^7$

tends to the Keating–Rudnick value 4 (4.35, 4.21 at $q = 16, 32$), unlike $d = 4$ where it tends to $3 \neq 2$. The value $q^{2(d/2-1)} = q^4$ suggested in [4] is excluded.

7.2. The geometry behind Y : a fixed nine-dimensional variety and a twisted Frobenius. We now give Y an unambiguous algebro-geometric meaning. The point is that Y_k is a Frobenius trace on a *fixed* variety over $\mathbb{F}_2$, but for a *twisted* Frobenius; in particular $(Y_k)_k$ is not the sequence of $\mathbb{F}_{2^k}$ -point counts of any single variety.

Definition 7.3. Let $z = (z_0, \dots, z_{11})$ be coordinates on $\mathbb{A}_{\mathbb{F}_2}^{12}$ and $e_m(z)$ the elementary symmetric polynomials. Put

$$I = \langle e_1, e_3, e_5 \rangle \subset \mathbb{F}_2[z], \quad X = V(I) \subset \mathbb{A}_{\mathbb{F}_2}^{12}, \quad f = e_4|_X, \quad \mathcal{L} = f^* \mathcal{L}_{\psi_0},$$

where $\mathcal{L}_{\psi_0}$ is the Artin–Schreier sheaf on $\mathbb{A}_{\mathbb{F}_2}^1$ attached to $\psi_0(x) = (-1)^x$, so that the trace function of $\mathcal{L} \otimes \mathbb{F}_q$ is $\psi_q \circ e_4$. Let $\sigma \in \text{Aut}(\mathbb{A}^{12})$ be the cyclic shift $\sigma(z)_j = z_{j-1}$ (indices mod 12); it is defined over $\mathbb{F}_2$, preserves I and e_4 , and commutes with the absolute Frobenius $F : z \mapsto (z_j^2)$. The σ -twisted $\mathbb{F}_q$ -form $X^{(\sigma)}$ of X is the $\mathbb{F}_q$ -structure whose geometric Frobenius is $\sigma \circ F^k$ (the class of a 12-cycle in $H^1(\mathbb{F}_q, S_{12})$; it is the twist defining $\text{Res}_{\mathbb{F}_{q^{12}}/\mathbb{F}_q} \mathbb{A}^1 \cong \mathbb{A}_{\mathbb{F}_q}^{12}$).

Proposition 7.4. (a) $I = \langle p_1, p_3, p_5 \rangle$ with $p_m = \sum_j z_j^m$ (Newton’s identities in characteristic 2: $p_3 = e_1^3 + e_1 e_2 + e_3$, $p_5 \equiv e_5 + e_2 e_3 \pmod{e_1}$).
 (b) $\alpha \mapsto (\alpha, \alpha^q, \dots, \alpha^{q^{11}})$ is a bijection $U \rightarrow X^{(\sigma)}(\mathbb{F}_q) = \{z \in X(\bar{\mathbb{F}}_2) : \sigma F^k(z) = z\}$ carrying $e_4(\alpha)$ to $e_4(z)$; hence $Y = \sum_{z \in X^{(\sigma)}(\mathbb{F}_q)} \psi_q(e_4(z))$.
 (c) Let $\pi : \mathbb{A}^{12} \rightarrow \mathbb{A}^{12}/S_{12} = \text{Spec } \mathbb{F}_2[e_1, \dots, e_{12}]$ and $\Lambda = V(e_1, e_3, e_5) \cong \mathbb{A}^9$ with coordinates $(e_2, e_4, e_6, e_7, \dots, e_{12})$. Then $X = \pi^{-1}(\Lambda)$, $\pi|_X : X \rightarrow \Lambda$ is finite flat of degree $12!$ and generically étale, $\Lambda(\mathbb{F}_q)$ is the set of monic $F \in \mathbb{F}_q[t]$ of degree 12 with $a_{11} = a_9 = a_7 = 0$, and $\#(\pi^{-1}(F) \cap X^{(\sigma)}(\mathbb{F}_q)) = \Lambda(F)$. Thus $Y = \sum_{F \in \Lambda(\mathbb{F}_q)} \Lambda(F) \psi_q(a_8(F))$, which is the Hayes-group form of Section 4; the projection $\Lambda \rightarrow \mathbb{A}^2$, (e_2, e_4) , has fibres of dimension 7.

Proof. (a) is immediate from Newton’s identities. (b): $\sigma F^k(z)_j = z_{j-1}^q$, so the fixed points are the tuples $z_j = z_0^{q^j}$ with $z_0^{q^{12}} = z_0$, and $e_1 = e_3 = e_5 = 0$ is $p_1 = p_3 = p_5 = 0$ by (a). (c): $\mathbb{F}_2[z]$ is free of rank $12!$ over $\mathbb{F}_2[e]$; the generic F is separable since $F' = e_7 t^4 + e_9 t^2 + e_{11} \neq 0$; the σF^k -fixed points of $\pi^{-1}(F)$ are the orderings $z_j = z_0^{q^j}$ of the roots of F , i.e. the $\alpha \in \mathbb{F}_{q^{12}}$ with $\text{charpoly}(\alpha) = F$, of which there are $\Lambda(F)$ (cf. (1)). $\square$

Proposition 7.5 (geometry of X). (a) X is a complete intersection of codimension 3 in $\mathbb{A}^{12}$ cut out by the regular sequence (e_1, e_3, e_5) of degrees 1, 3, 5; it is an affine cone of pure Krull dimension 9 and is Cohen–Macaulay. (“Nine-dimensional” refers to $\dim X$; the fibres of $X \rightarrow \Lambda$ have dimension 0 and those of $X \rightarrow \mathbb{A}^2$ dimension 7.)

- (b) $\text{Sing}(X) = X \cap \{z : \#\{z_0, \dots, z_{11}\} \leq 2\}$ is the union of the 1023 planes $P_A = \{z_j = u \ (j \in A), z_j = v \ (j \notin A)\}$, $A \subset \{0, \dots, 11\}$ of even cardinality $0 < |A| < 12$ (A and its complement give the same plane), all containing the diagonal line; $\dim \text{Sing}(X) = 2$, of codimension 7.
 (c) X is geometrically integral and normal. Consequently the geometric monodromy group of the family $\{F : e_1 = e_3 = e_5 = 0\}$, i.e. the Galois group of $t^{12} + e_2 t^{10} + \dots + e_{11} t + e_{12}$ over $\bar{\mathbb{F}}_2(e_2, e_4, e_6, \dots, e_{12})$, is the full S_{12} .

Proof. (a) By Proposition 7.4(c), $\mathbb{F}_2[z]/I$ is a free module over the ring $\mathbb{F}_2[e]/(e_1, e_3, e_5)$, which is the polynomial ring $\mathbb{F}_2[e_2, e_4, e_6, \dots, e_{12}]$; so (e_1, e_3, e_5) is a regular sequence and X is finite over $\mathbb{A}^9$. (b) By the Jacobian criterion for complete intersections, X is smooth at z iff de_1, de_3, de_5 are independent there. On X , $\partial_j e_1 = 1$, $\partial_j e_3 = e_2(z \setminus z_j) = e_2 + z_j^2$ and $\partial_j e_5 = e_4(z \setminus z_j) = e_4 + e_2 z_j^2 + z_j^4$, so the row space is that of the Vandermonde rows $(1)_j, (z_j^2)_j, (z_j^4)_j$, of rank 3 iff at least three of the z_j are distinct. Conversely, if z takes only the values u, v with multiplicities s, r , $s + r = 12$, then $e_m(z) = \sum_i \binom{s}{m-i} \binom{r}{i} u^{m-i} v^i$; for r even every term with m odd contains a factor $\binom{\text{even}}{\text{odd}}$, which is even by Lucas’ theorem, so $e_1 = e_3 = e_5 = 0$ and $P_A \subset X$; for r odd,

$e_1 = u + v$ forces $u = v$. (c) X is generically reduced (generic fibre of π étale) and Cohen–Macaulay, hence reduced; Cohen–Macaulay gives Serre’s S_2 and (b) gives R_1 , so X is normal; a cone is connected, and a connected normal variety is irreducible. The same holds over $\mathbb{F}_2$. The irreducible components of $X_{\mathbb{F}_2}$ correspond to the cosets S_{12}/G of the geometric monodromy group G (the generic fibre of π is the splitting algebra), so $G = S_{12}$. $\square$

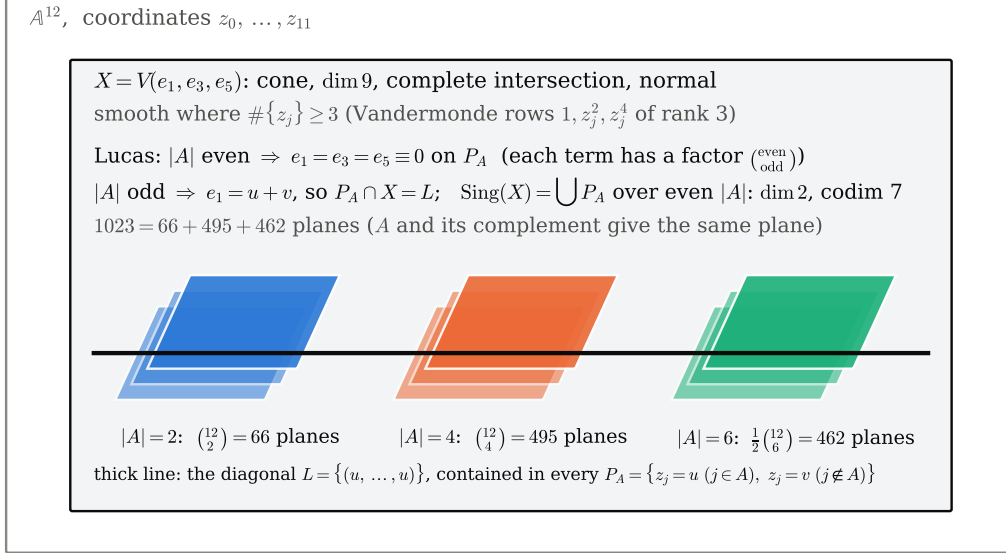


FIGURE 4. The singular locus of $X = V(e_1, e_3, e_5) \subset \mathbb{A}^{12}$ (Proposition 7.5(b)), schematically. A point with only two distinct coordinate values u, v lies on the plane P_A determined by the index set A of the coordinates equal to u ; by Lucas’ theorem $P_A \subset X$ exactly when $|A|$ is even, and the 1023 such planes, stratified by $|A| \in \{2, 4, 6\}$ (A and its complement give the same plane), all contain the diagonal line and form the two-dimensional singular locus, of codimension 7 in X . Outside these planes at least three coordinates are distinct and the Vandermonde rows $(1), (z_j^2), (z_j^4)$ make X smooth.

Proposition 7.6 (the Artin–Schreier sheaf). *The cover $Z = \{(z, y) \in X \times \mathbb{A}^1 : y^2 + y = e_4(z)\} \rightarrow X$ is étale of degree 2 and Z is smooth over X_{sm} . The function $f = e_4$ has no poles on X ; on the projective closure $\overline{X} \subset \mathbb{P}^{12}$ its polar divisor is $4X_\infty$ with $X_\infty \cong \mathbb{P}(X) \subset \mathbb{P}^{11}$ the base of the cone. Since the pole order 4 is divisible by $p = 2$, the Swan conductor of $\mathcal{L}$ along X_∞ is < 4 . The sheaf $\mathcal{L}$ is geometrically nontrivial on X (on the diagonal line $e_4 = \binom{12}{4}u^4 = u^4$, and $u^4 \notin \{g^2 + g + c\}$), so Z is geometrically connected and*

$$H_c^{18}(X_{\mathbb{F}_2}, \mathcal{L}) = H^0(X_{\mathbb{F}_2}, \mathcal{L}^\vee)^\vee(-9) = 0.$$

Theorem 7.7 (twisted trace formula). *For every $k \geq 1$, with F the geometric Frobenius of $\mathbb{F}_2$ acting on ℓ -adic cohomology ($\ell \neq 2$) and σ acting through its lift to $\mathcal{L}$,*

$$Y_k = \sum_{j=0}^{18} (-1)^j \text{Tr}(\sigma \circ F^k \mid H_c^j(X_{\mathbb{F}_2}, \mathcal{L})).$$

Consequently $\sum_{k \geq 1} Y_k T^k = T \frac{d}{dT} \log L_\sigma(T)$ with $L_\sigma(T) = \prod_j \det(1 - \sigma F T \mid H_c^j)^{(-1)^{j+1}} \in \mathbb{Q}(T)$; $(Y_k)_k$ is a linear recurrence sequence, and $m_k = Y_k/8^k = \sum_i \pm \zeta_i \eta_i^k$ with $\zeta_i^{12} = 1$ and $\eta_i = \lambda_i/8$, λ_i the eigenvalues of F . By Deligne [6] the λ_i on H_c^j are algebraic integers of weight $\leq j$; the

normalisation $8^k = q^3$ is the Tate twist by 3, so $H_c^j(3)$ has weights $\leq j - 6$. If the cohomology were concentrated in degree $9 = \dim X$ and pure, then $m_k = \pm \sum_{i=1}^{b_9} \zeta_i \eta_i^k$ with $|\eta_i| = 2^{3/2}$ and $|m_k| \leq b_9 q^{3/2}$.

Proof. By Proposition 7.4(b), Y_k is the sum of the trace function of $\mathcal{L}$ over the $\mathbb{F}_q$ -points of the twisted form $X^{(\sigma)}$, whose geometric Frobenius acts on $H_c^\bullet(X_{\mathbb{F}_2}, \mathcal{L})$ as σF^k (σ has finite order and commutes with F); apply the Grothendieck–Lefschetz formula (SGA 4 $\frac{1}{2}$, Rapport; for $F \circ \sigma$ with σ of finite order see also Deligne–Lusztig [7, proof of Thm. 3.2]). Since σ and F commute they are simultaneously triangularisable, which gives the stated shape; L_σ has coefficients in $\mathbb{Q}(\zeta_{12})$ a priori and in $\mathbb{Q}$ because all $Y_k \in \mathbb{Z}$. The weight statements are Weil II. $\square$

By contrast with A_6 and X , the function $(-1)^{\text{Tr } e_4}$ on $\mathcal{G}^2$ is not a character: in Witt coordinates $\mathcal{G}^2 \cong W_2(\mathbb{F}_q)$, $(x_1, x_2) + (x'_1, x'_2) = (x_1 + x'_1, x_2 + x'_2 + x_1 x'_1)$, and the carry destroys multiplicativity (the characters of order 4 are $i^{\text{Tr } x_1} (-1)^{\text{Tr } x_2 + \dots}$). Therefore Y does not decompose into Frobenius traces of individual L -functions of curves and no supersingularity argument applies; it is the global twisted trace on the nine-dimensional X .

Proposition 7.8. *The values $m_k = 13, -1, -35, 47, 733$ ($k = 1, \dots, 5$) satisfy $|m_k|/q^{3/2} = 4.60, 0.13, 1.55, 0.73, 4.05$. Hence, in the pure, degree-9 situation of Theorem 7.7, $b_9 \geq 5$. The elementary symmetric functions of the $\zeta_i \eta_i$ computed from $m_1, \dots, m_5$ by Newton’s identities are 13, 85, 361, 1069, 2281; if $b_9 = 5$ then $|e_5| = \prod |\eta_i| = 2^{15/2} \neq 2281$, a contradiction (the argument uses only absolute values, so the twists ζ_i do not affect it). Thus either $b_9 \geq 6$, or $H_c^\bullet$ has contributions outside the pure weight-9 part.*

Proposition 7.4(b) was checked directly: enumerating $X^{(\sigma)}(\mathbb{F}_2)$ inside $\mathbb{F}_{2^{12}}$ gives $|U| = 544$, $Y_1 = 104$, and the pushforward formula of (c) gives $(|U|, Y) = (544, 104)$ and $(259072, -64)$ for $q = 2, 4$ (script `verify_Y_direct.py`; a SageMath version is `verify_Y_sage.sage`).

Remark 7.9 (what is and is not established). Theorem 7.7 and Propositions 7.4–7.6 are unconditional. We have *not* shown that $H_c^j(X_{\mathbb{F}_2}, \mathcal{L})$ vanishes for $9 < j < 18$ (Proposition 7.6 gives only $j = 18$), nor computed b_9 : Katz’s theorem on the concentration of $\sum \psi(f)$ in middle degree requires $p \nmid \deg f$ and a smooth ambient variety, whereas here $\deg e_4 = 4$ and X is singular along the 1023 planes of Proposition 7.5(b); a proof would need the Swan conductor of $\mathcal{L}$ along X_∞ and local computations at $\text{Sing}(X)$. Hence $m_k = O(q^{3/2})$, and with it $C_6 = 1$, is supported by the data but not proved. Two further cautions. First, the variety $V = \{p_1 = p_3 = p_5 = 0\} \subset \mathbb{A}_{\mathbb{F}_q}^{12}$ obtained by choosing an $\mathbb{F}_q$ -basis of $\mathbb{F}_{q^{12}}$ is exactly $X^{(\sigma)}$, a nontrivial twist of $X \otimes \mathbb{F}_q$; so $(Y_k)_k$ is not a point-count sequence of a fixed variety, and the genuine “closed form” of Y is the twisted L -function $L_\sigma(T)$, which Theorem 7.7 shows is a rational function determined by finitely many Y_k (Berlekamp–Massey on $Y_1, Y_2, \dots$ would identify it; $k \leq 5$ is not enough and $k = 6$ means $\mathbb{F}_{2^{72}}$). Second, the untwisted counts $\#X(\mathbb{F}_q)$ for $q = 2, 4, 8$ (2048, 2 099 200, 1 081 081 856) far exceed q^9 ; this is not a sign of several components but of $q < 12$, where no polynomial over $\mathbb{F}_q$ has twelve distinct roots. The tower reduction of Theorem 5.3 exhibits $Y/G - 1$ as a sum over $a \in \mathbb{F}_{q^6}$ of cubic Artin–Schreier sums over $\mathbb{F}_q$, i.e. as the trace of a relative H^1 of an elliptic Artin–Schreier family over a five-dimensional base; this is the natural starting point for computing b_9 .

8. SUMMARY OF THE NUMERICAL EVIDENCE

REFERENCES

- [1] B. C. Berndt, R. J. Evans and K. S. Williams, *Gauss and Jacobi Sums*, Wiley, New York, 1998.
- [2] A. W. Bluher, *On $x^{q+1} + ax + b$* , Finite Fields Appl. **10** (2004), 285–305.
- [3] L. Carlitz, *Explicit evaluation of certain exponential sums*, Math. Scand. **44** (1979), 5–16.
- [4] R. Chen, *Primes in Legendre intervals over $\mathbb{F}_q[t]$: an exact explicit-formula algorithm, verification for $d \leq 10$, and an anomaly in characteristic 2*, preprint (2026), DOI 10.5281/zenodo.23219609.
- [5] R. S. Coulter, *On the evaluation of a class of Weil sums in characteristic 2*, New Zealand J. Math. **28** (1999), 171–184.
- [6] P. Deligne, *La conjecture de Weil. II*, Publ. Math. IHES **52** (1980), 137–252.

TABLE 4. Status of the main identities. “exact” means verified by exact integer computation; the method is indicated.

Statement	Status	Range checked
$d = 4$: Theorem 3.2 (all classes)	proved	formula = FFT class by class, $q \leq 128$
$d = 4$: $\text{Var}_{\mathcal{G}^2}/q^5 = q(q-1)$, $\text{Var}_{\mathcal{G}}/q^5 = 3 - 3/q$	proved	Lemma 3.10 also checked to $q = 2^{22}$
$d = 4$: ρ structure (Theorem 3.12)	proved	matches [4, Table 4]
$d = 6$: three-value profile (Theorem 4.2)	proved	all q^2 classes, $q \leq 32$
$d = 6$: tower identities and reductions	proved	$\mathbb{F}_{2^{12}}$ exhaustive; $q = 2, 4, 8$
$d = 6$: pure cubic and pure quintic parts (Lemmas 6.2, 6.4)	proved, all k	exact, $q \leq 128$
$d = 6$: mixed part, k odd (Theorem 6.12); types of $v^5 + v + c$ (Proposition 6.8)	proved	quadratic forms $k \leq 7$; L -polynomials $k \leq 9$; types $k \leq 11$
$d = 6$: $A_6 = q^9 + (q-1)q^5$, k odd (Corollary 6.13)	proved	exact, $q \leq 128$ (Table 1)
$d = 6$: A_6 for even k and X for all k (Conjecture 6.5)	open	exact, $q \leq 128$ (Table 1)
$d = 6$: $\text{Var}_{\mathcal{G}^2}/q^7$ formula (Theorem 7.1)	proved given Conj. 6.5 (X-part only for odd k)	$q \leq 32$
$C_6 = 1$	conditional on $m_k = o(q^{5/2})$	$q \leq 32$

- [7] P. Deligne and G. Lusztig, *Representations of reductive groups over finite fields*, Ann. of Math. (2) **103** (1976), 103–161.
- [8] G. van der Geer and M. van der Vlugt, *Reed–Muller codes and supersingular curves. I*, Compositio Math. **84** (1992), 333–367.
- [9] T. Helleseht and A. Kholosha, *On the equation $x^{2^l+1} + x + a = 0$ over $GF(2^k)$* , Finite Fields Appl. **14** (2008), 159–176.
- [10] H. Hasse and H. Davenport, *Die Nullstellen der Kongruenzzetafunktionen in gewissen zyklischen Fällen*, J. reine angew. Math. **172** (1935), 151–182.
- [11] N. M. Katz, *Witt vectors and a question of Keating and Rudnick*, Int. Math. Res. Not. IMRN (2013), 3613–3638.
- [12] J. P. Keating and Z. Rudnick, *The variance of the number of prime polynomials in short intervals and in residue classes*, Int. Math. Res. Not. IMRN (2014), 259–288.
- [13] J. Lahtonen, G. McGuire and H. N. Ward, *Gold and Kasami–Welch functions, quadratic forms, and bent functions*, Adv. Math. Commun. **1** (2007), 243–250.

GUT GEOSERVICE INC., MONTREAL, CANADA
 Email address: ruqing@hotmail.com